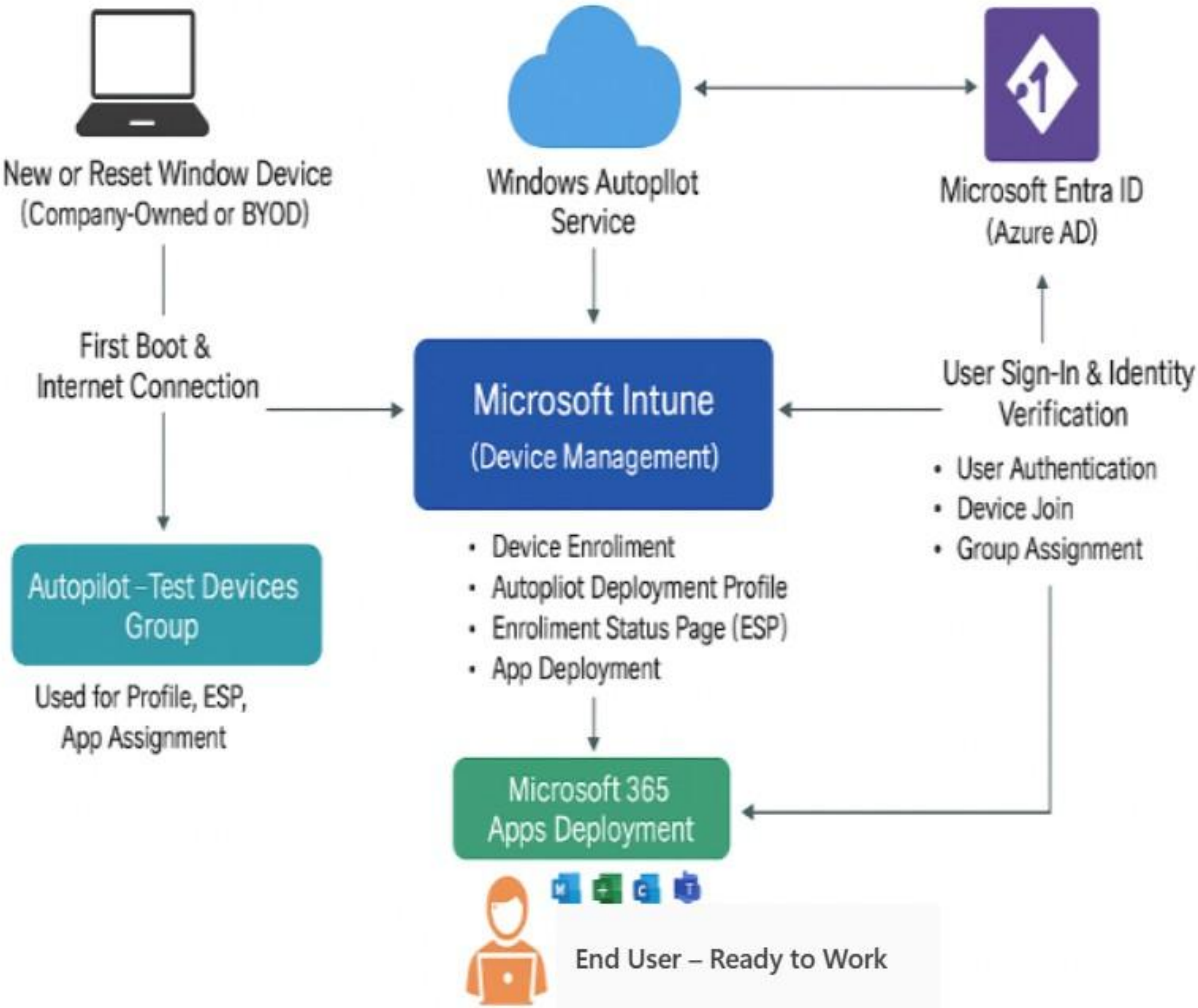
Windows Autopilot – End-to-End Device Enrollment

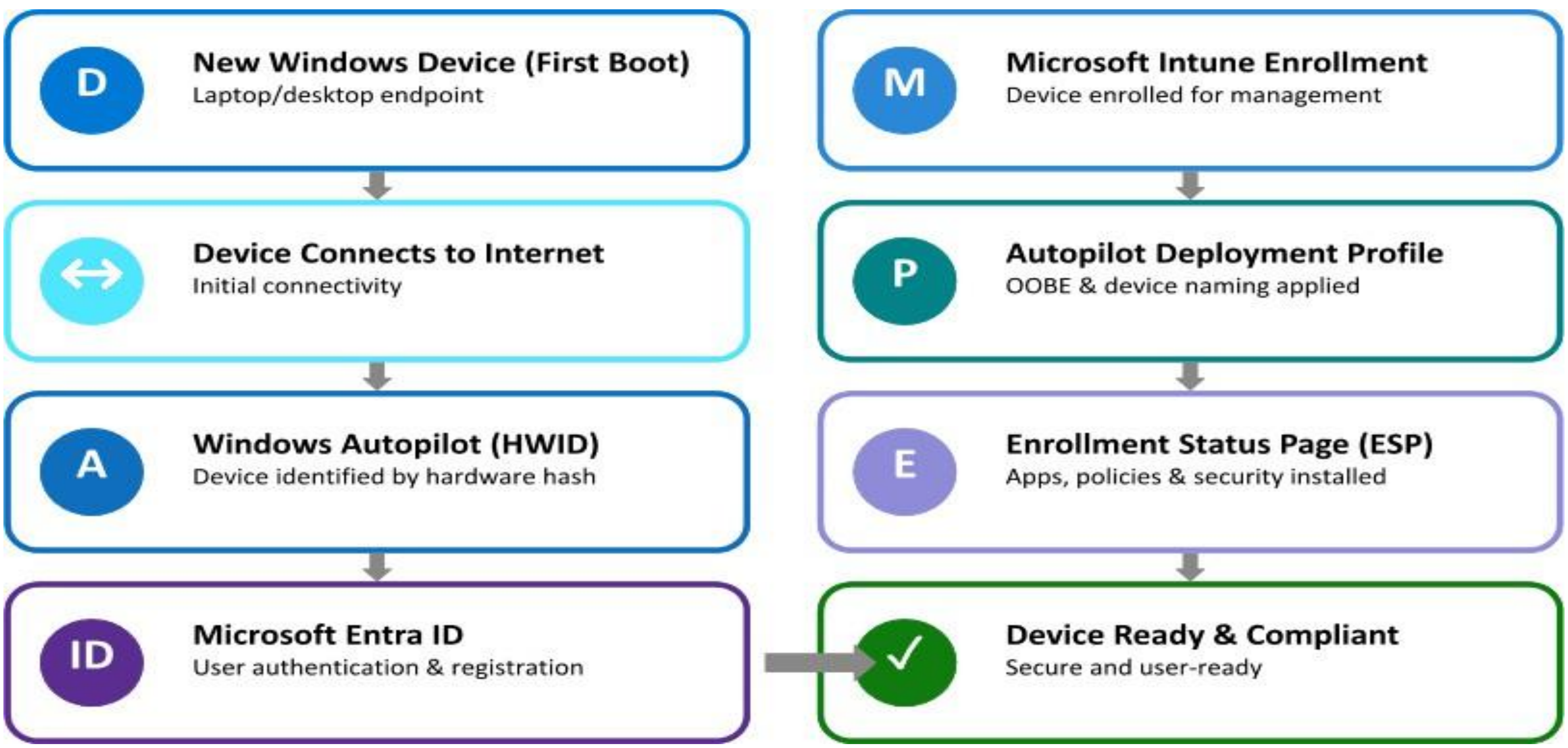


Streamlining Windows Setup and Software Delivery Using Autopilot

❖ What is Windows Autopilot?

- Windows Autopilot is a cloud-based provisioning technology that makes setting up new Windows devices super easy. Instead of IT staff manually installing apps and settings, Autopilot lets you prepare everything online. When someone turns on a new device and signs in, it automatically connects to the cloud, provisions itself with the right settings, apps, and security rules, and becomes ready to use no tech support needed. It's like handing users a brand-new, work-ready computer straight out of the box.

- Windows Autopilot improves security by ensuring every device is set up with the correct policies, configurations, and security settings right from the start. Since provisioning happens through the cloud, IT can enforce encryption, antivirus, firewall rules, and compliance policies automatically. Devices are also joined to Azure Active Directory and enrolled in Intune, which helps manage them securely and consistently reducing risks from manual setup errors or unauthorized changes.



❖ Autopilot: Streamlining Device Configuration and Protection for Today's Distributed Teams

- Traditionally, companies buy laptops from vendors like Dell and manually set them up before handing them to employees. But with the rise of Bring Your Own Device (BYOD) and remote work, this method has become outdated. For instance, if Microsoft orders 1000 laptops, Dell can supply the hardware and a list of device IDs, but it can't apply Microsoft's internal settings or security standards.

- This is where Windows Autopilot steps in. Using cloud-based provisioning and Microsoft Intune, companies can register devices with their hardware ID. Whether the device is company-issued or personally owned, once it connects to the internet, Intune automatically identifies it and applies the correct apps, settings, and security policies. This ensures every device no matter where it's located is securely configured and ready to use.

- For remote workers, this is especially valuable. Devices can be shipped directly to employees and set up without IT involvement. If a device is lost or stolen, Intune can remotely lock or wipe it to protect sensitive data. Unlike traditional imaging, Autopilot provides a pre-configured experience from the first boot, making device deployment faster, safer, and more efficient for modern organizations.

❖ Objectives of Windows Autopilot

- Automate Device Enrollment: Simplify and streamline the process of registering devices into Microsoft Intune without manual IT intervention.

- Enable Zero-Touch Provisioning: Allow devices to be configured automatically with company-specific settings, apps, and policies as soon as they connect to the internet.

- Support BYOD and Remote Work: Ensure secure and consistent setup for both company-owned and personal devices, regardless of location.

- Enhance Security and Compliance: Apply security configurations like encryption, antivirus, and firewall rules automatically to protect data and meet compliance standards.

- Reduce IT Workload: Minimize the need for hands-on setup, imaging, and manual configuration by using cloud-based provisioning.

- Improve User Experience: Deliver a ready-to-use device experience from the first boot, making onboarding faster and smoother for employees.

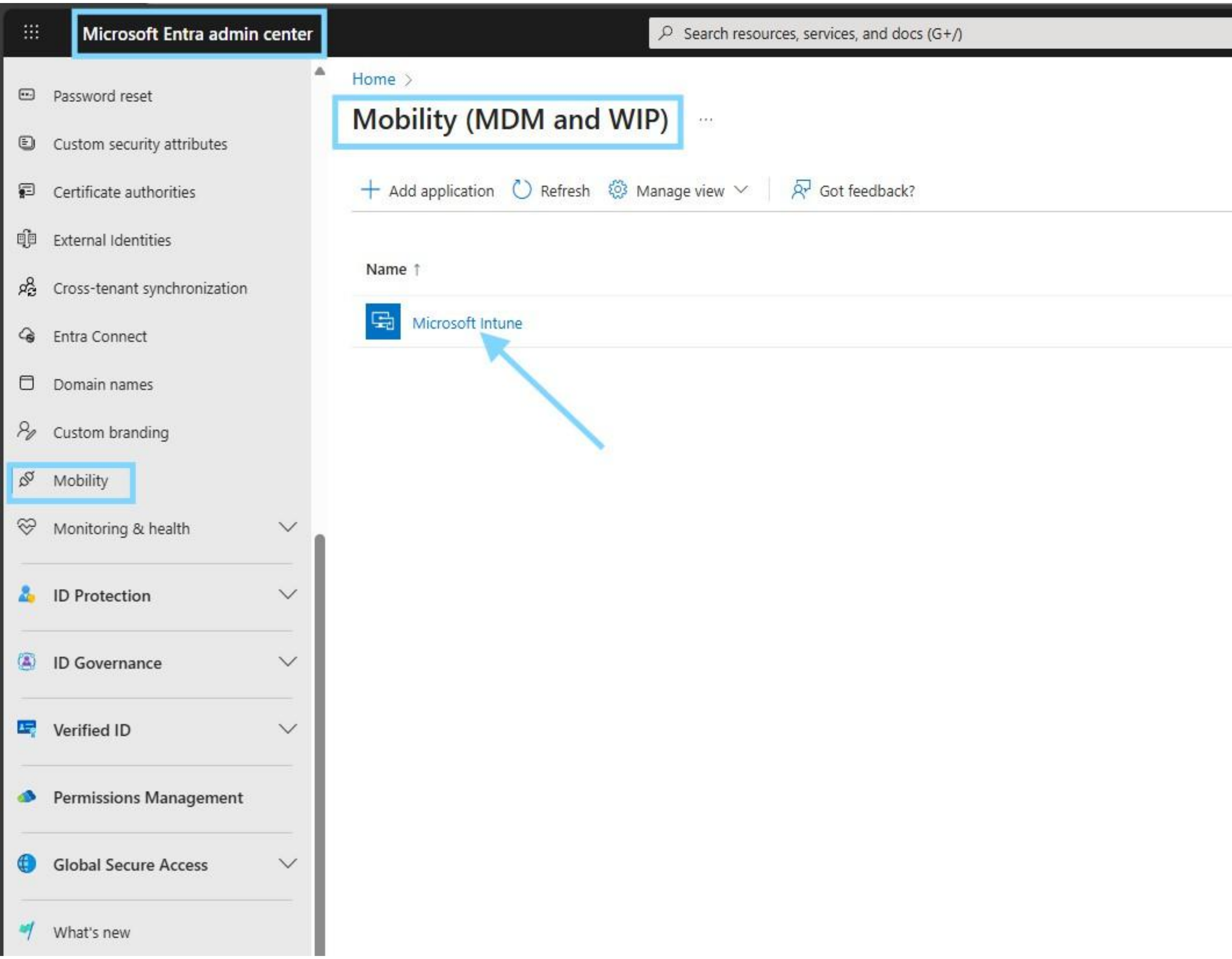
Windows Autopilot Device Enrollment – Overview

This diagram provides a high-level overview of the **Windows Autopilot device enrollment process** using **Microsoft Entra ID** and **Microsoft Intune**. It illustrates how a new Windows device is automatically configured and prepared for end-user use without manual IT intervention.

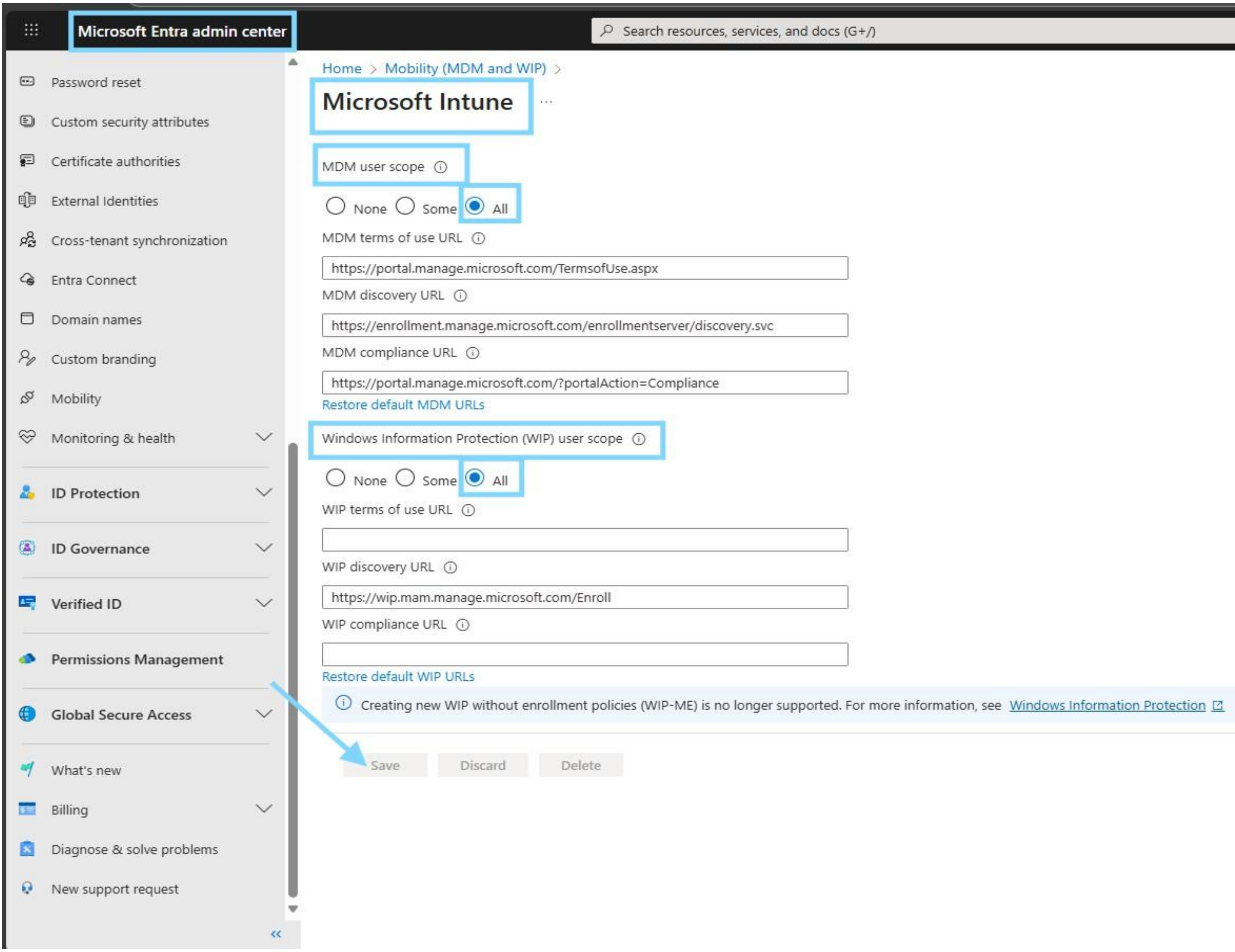
The process begins when a new or reset device is powered on and connected to the internet. Windows Autopilot identifies the device using its **hardware ID (HWID)** and initiates the enrollment process. Microsoft Entra ID handles user authentication and device registration, while Microsoft Intune enrolls the device for management.

During enrollment, Intune applies the **Autopilot deployment profile**, installs required applications, and enforces security and compliance policies through the **Enrollment Status Page (ESP)**. Once all configurations are completed successfully, the device becomes **secure, compliant, and ready for the user**.

Go to Microsoft Entra admin center.

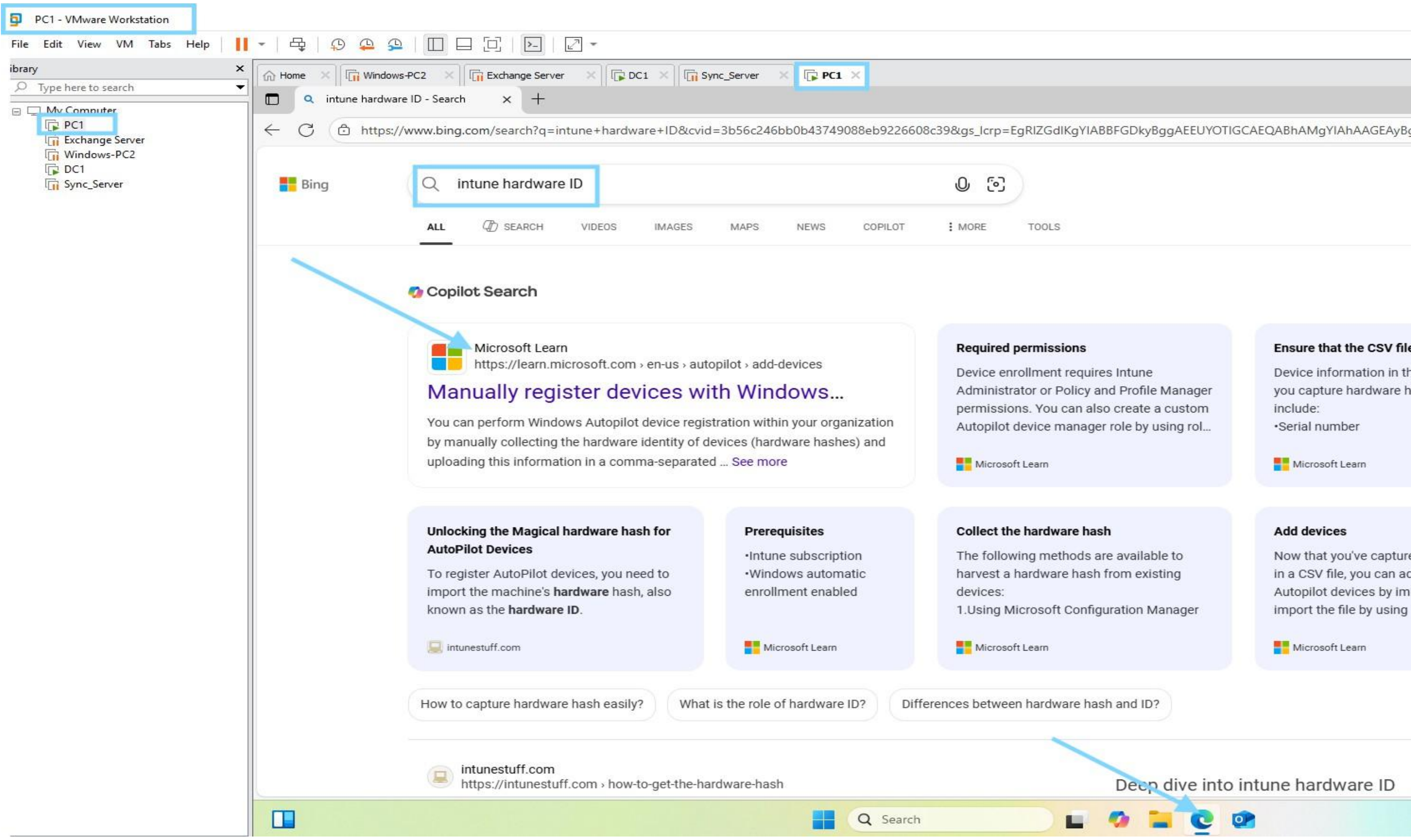


Enabling the MDM user scope sets Microsoft Intune as the device management authority so all users' Windows devices are automatically enrolled and managed during Autopilot setup.

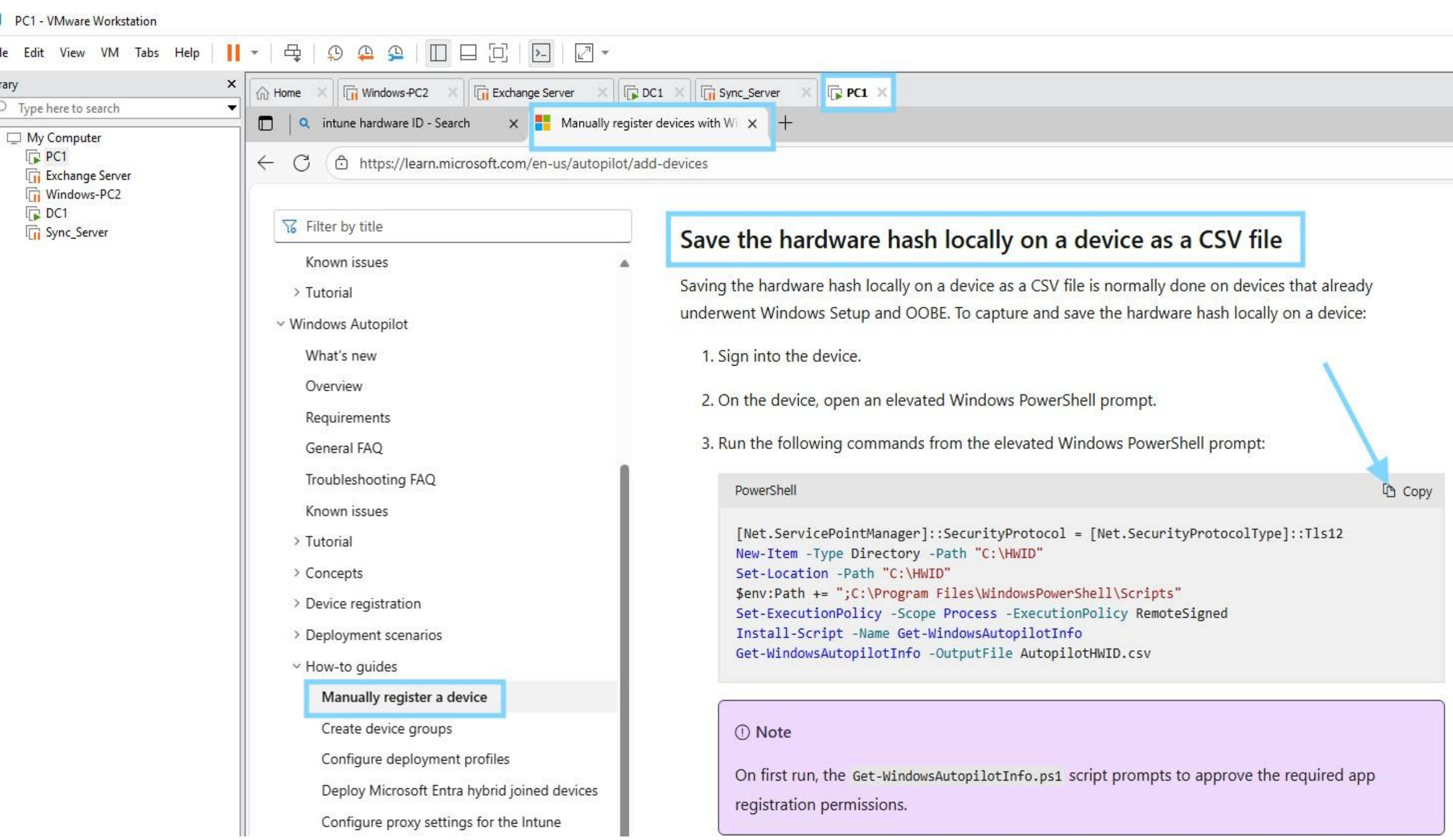


To generate and upload hardware IDs, you first need to collect them from your devices. This can be done using tools like PowerShell. In most cases, manufacturers provide these identifiers in a CSV file, which you can then import into Intune for Autopilot enrollment. Since you're working on VMware, you can run the PowerShell script inside the virtual machine to capture the hardware hash.

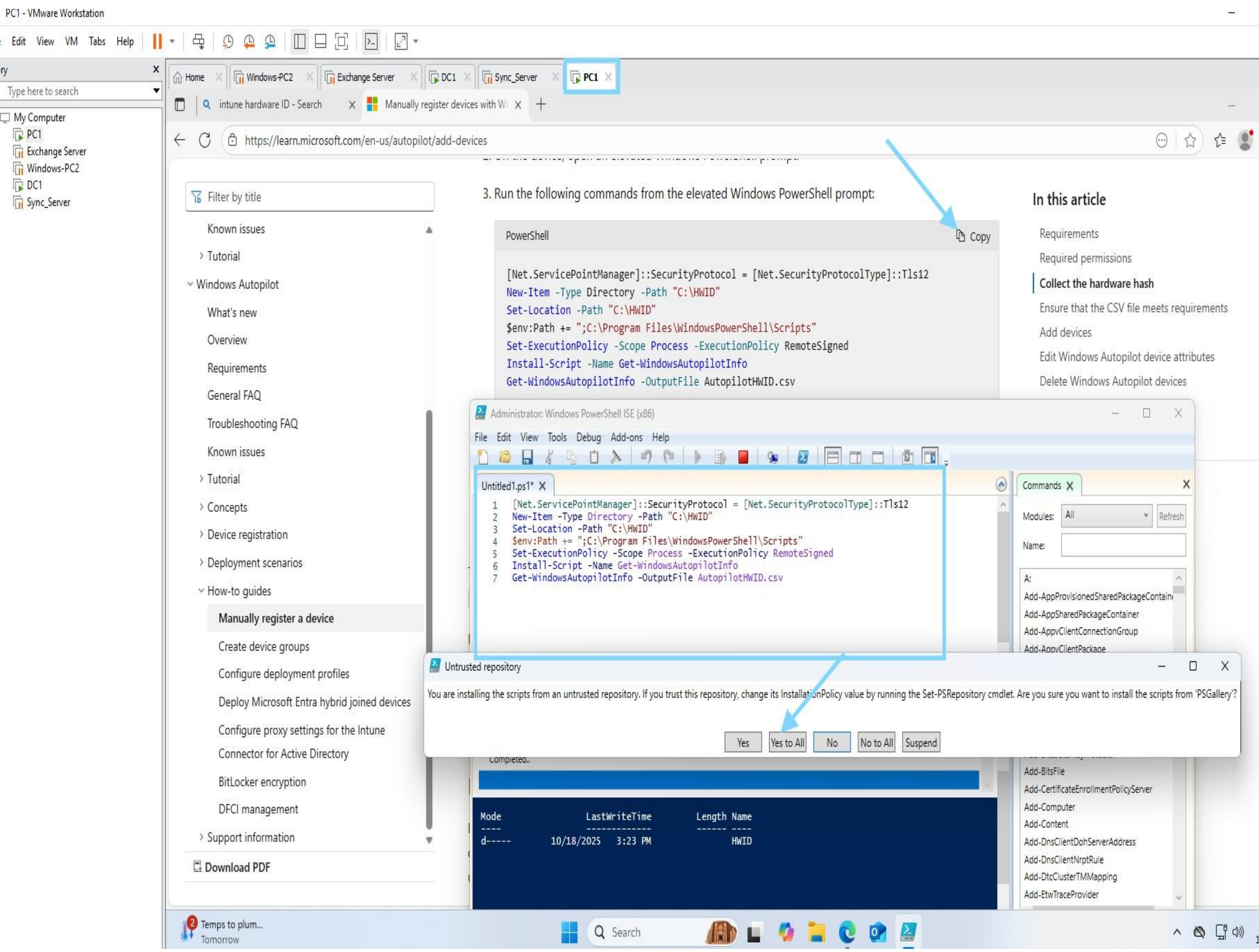
The hardware ID is captured so the device can be registered in Windows Autopilot and automatically enrolled into Intune.



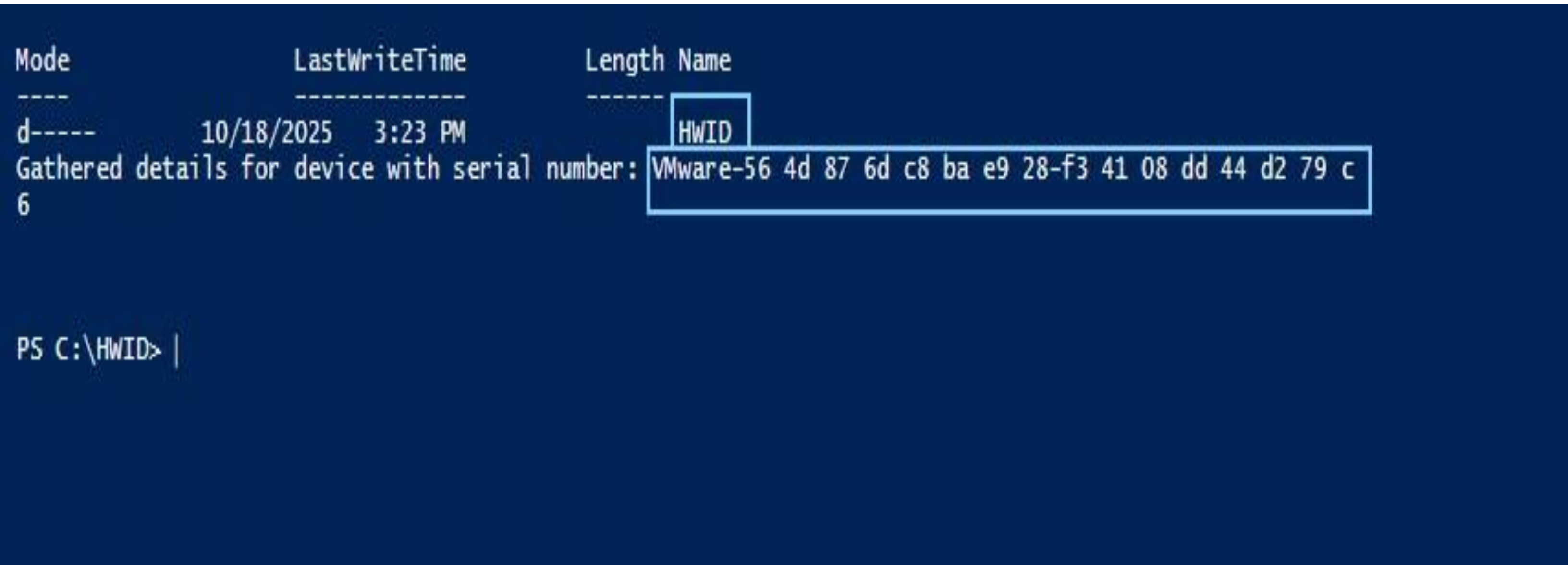
The hardware hash is collected and saved locally to register the device with Windows Autopilot.



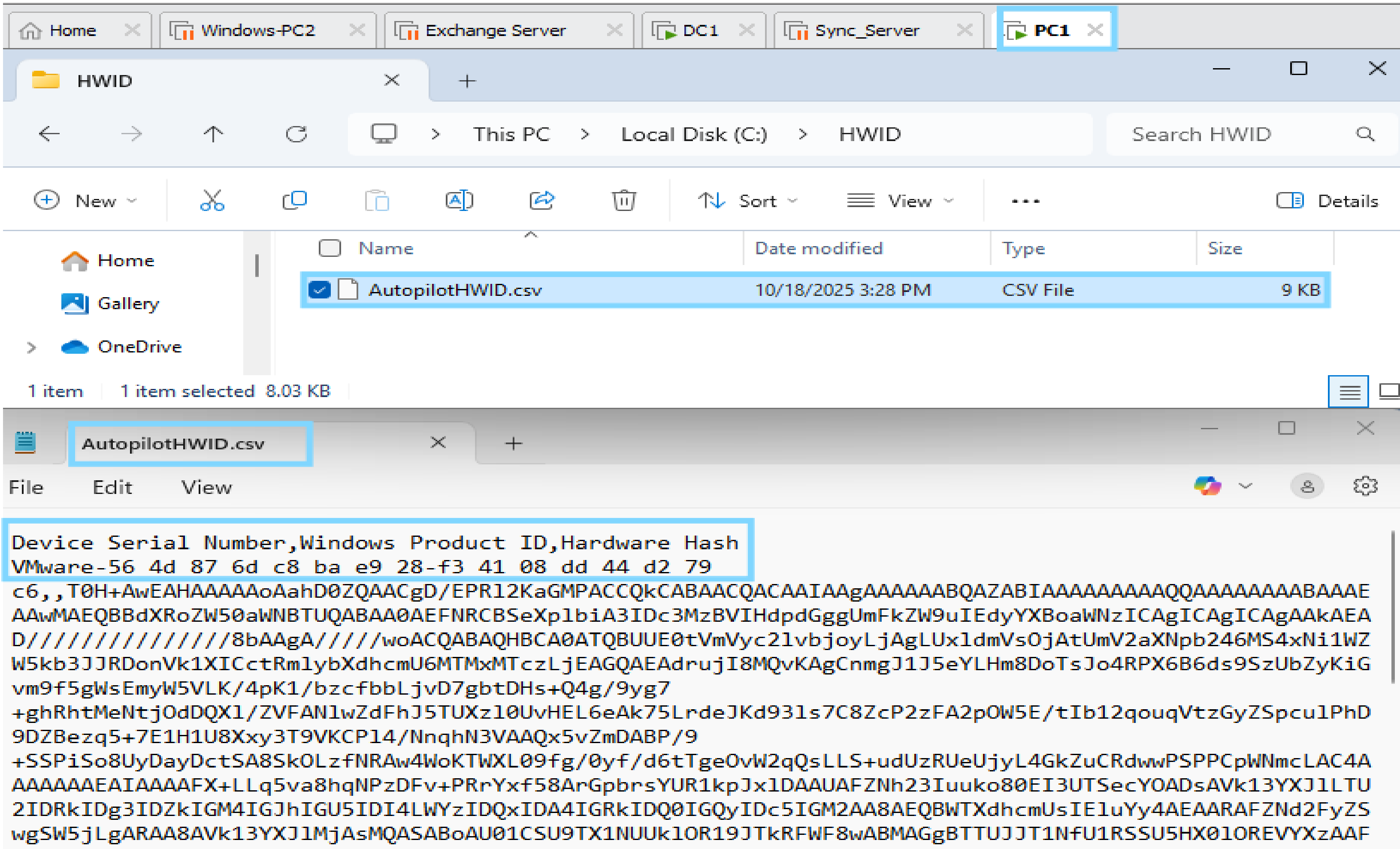
An elevated PowerShell script is executed to collect the device hardware hash and generate a CSV file for Windows Autopilot registration.

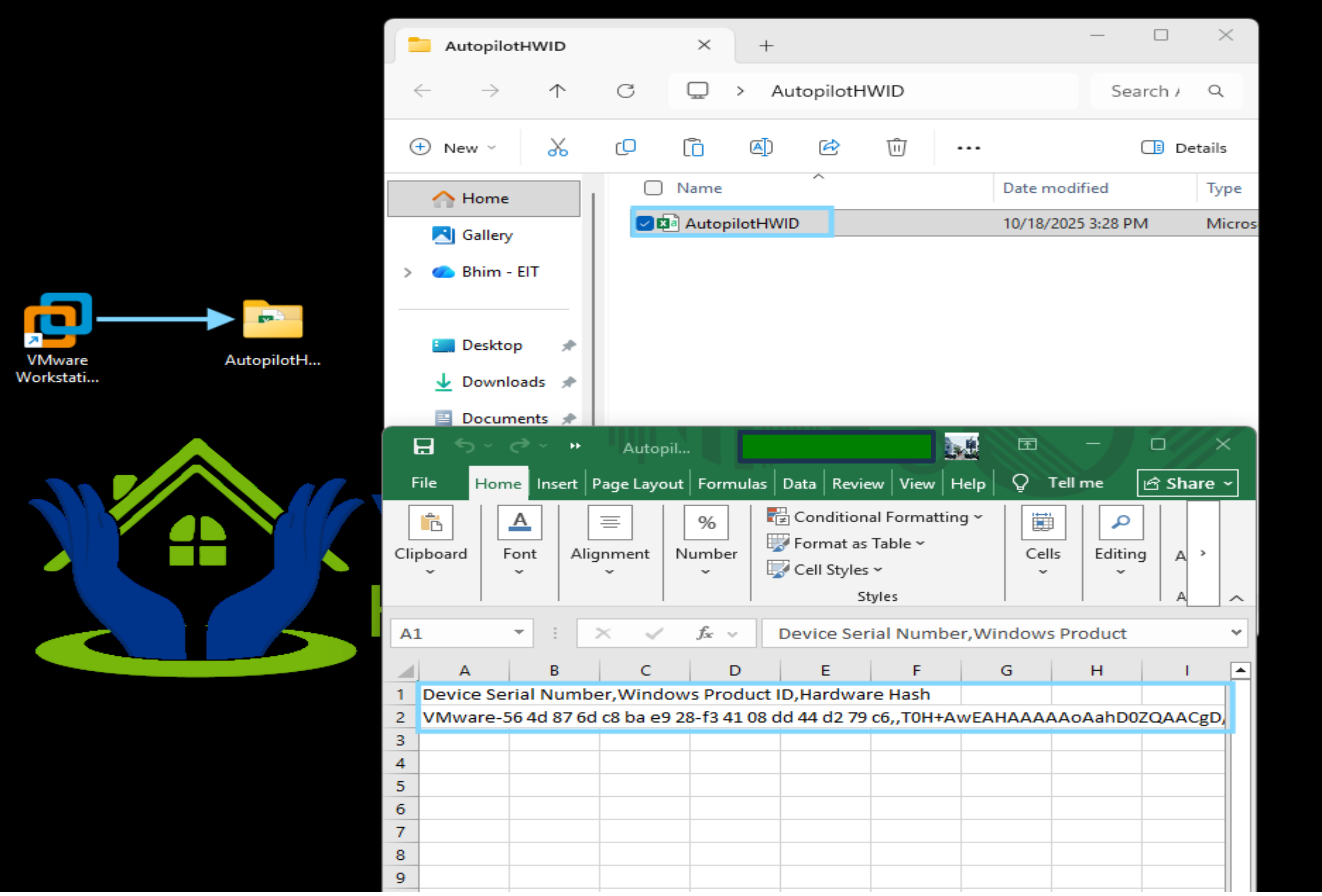


The device hardware ID is successfully generated, confirming readiness for Windows Autopilot enrollment.

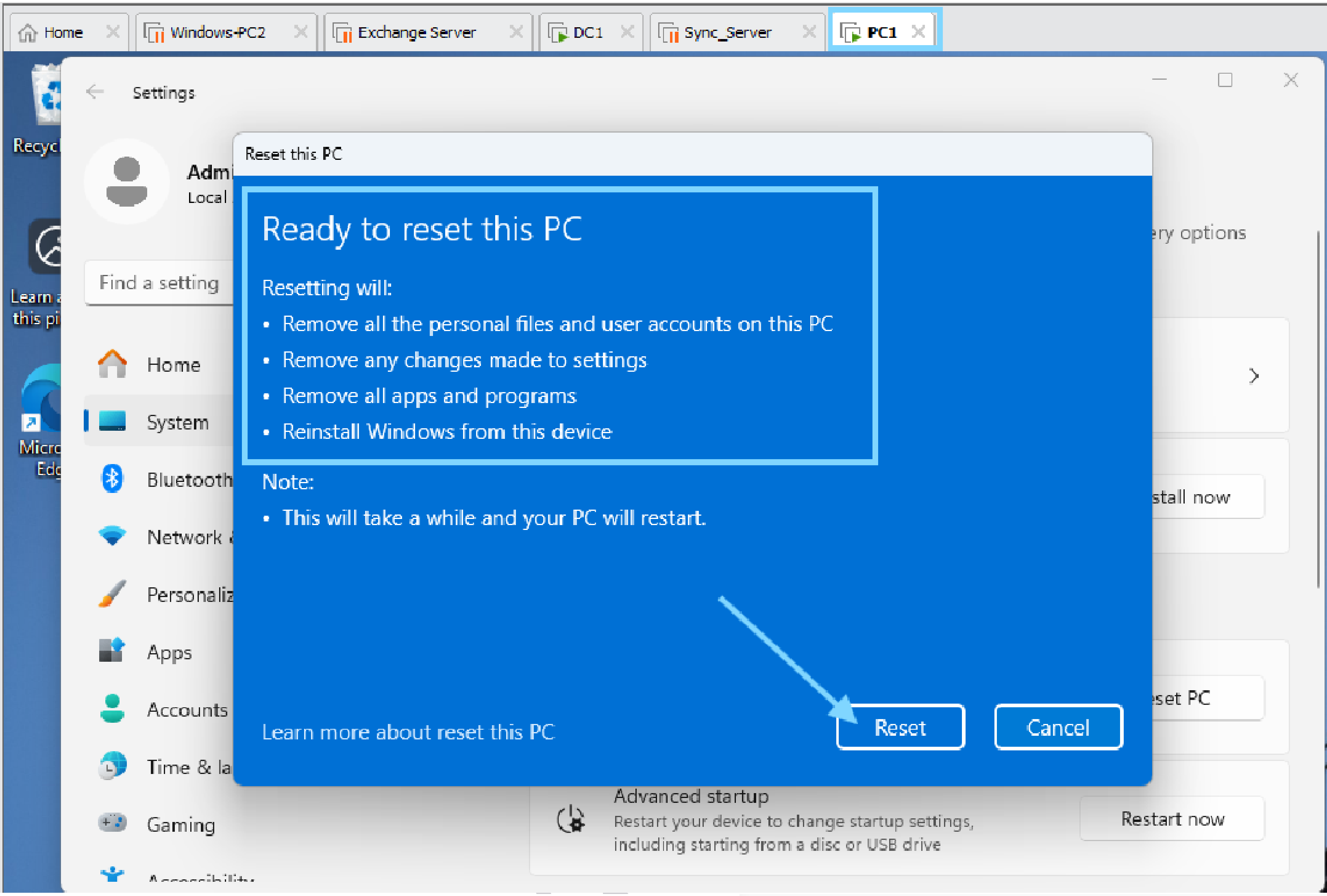


The device hardware hash is saved as a CSV file, enabling registration with Windows Autopilot through Intune.





Reset settings are finalized to ensure a clean OS state for Windows Autopilot deployment.



Configuring MDM Enrollment and Licensing

MDM Enrollment Configuration

Configure Intune as the MDM authority and enable automatic enrollment for applicable devices in Autopilot.

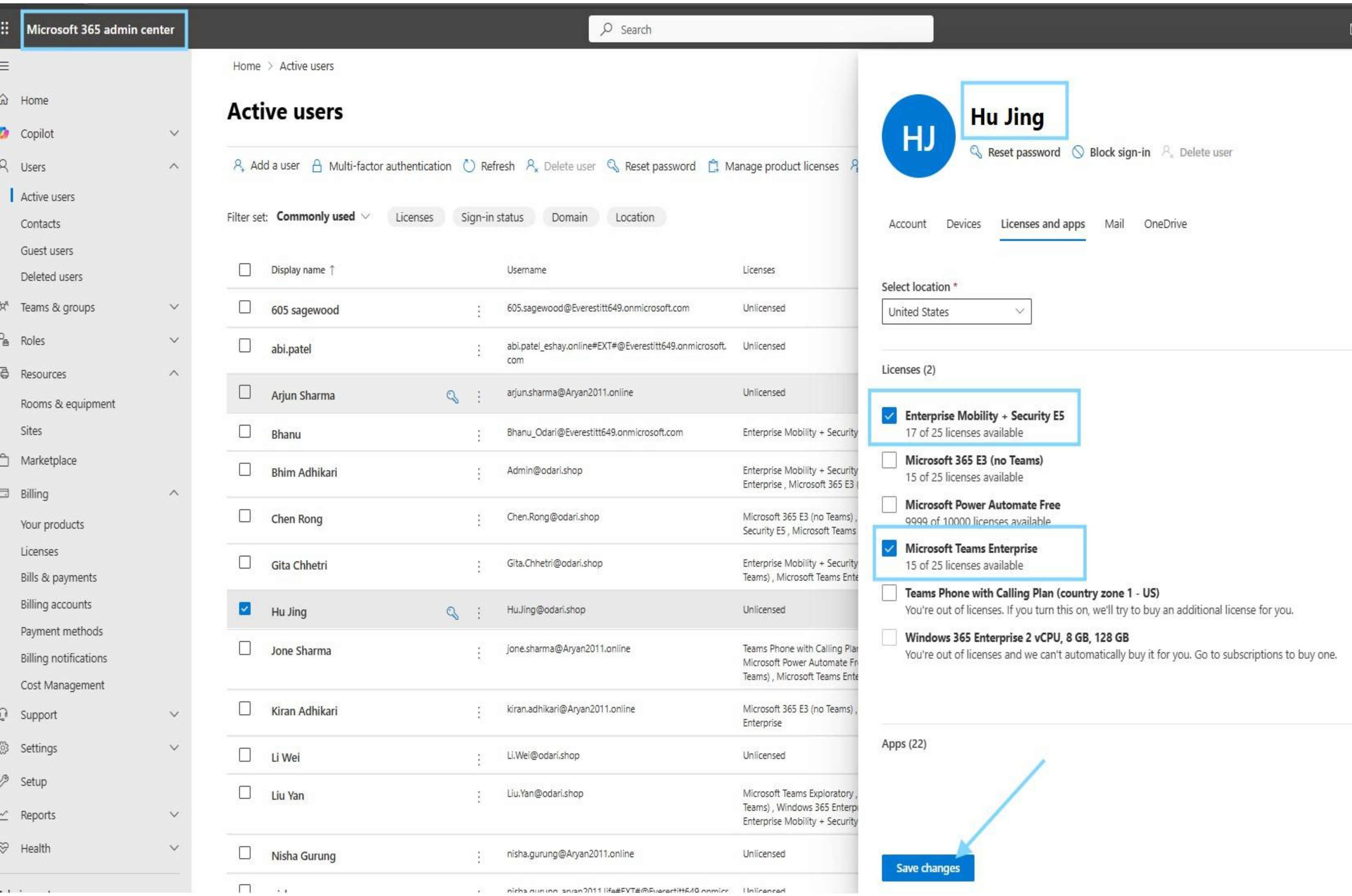
Licensing Importance

Assign valid Intune and Microsoft 365 licenses to users to enable provisioning and app deployment access.

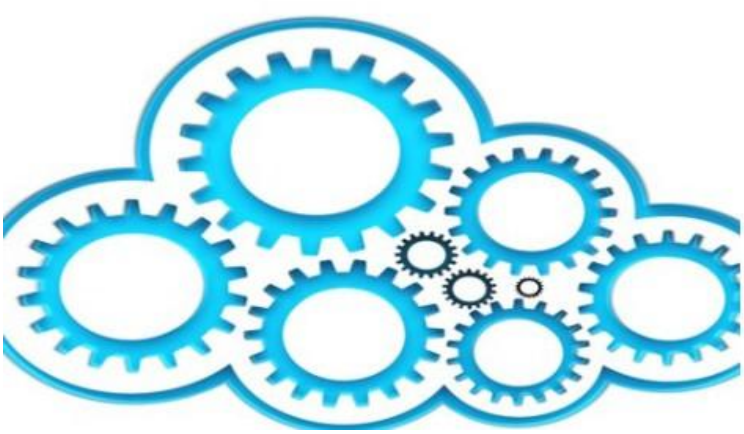
User Group and Policy Scoping

Verify test user membership in Autopilot-Test Devices group to apply correct policies and profiles during setup.

Appropriate Microsoft 365 and Intune licenses are assigned to the user to support Windows Autopilot deployment.



Creating Device Groups for Autopilot



Role of Device Groups

Device groups in Azure AD assign deployment profiles and manage Enrollment Status Page configurations effectively.

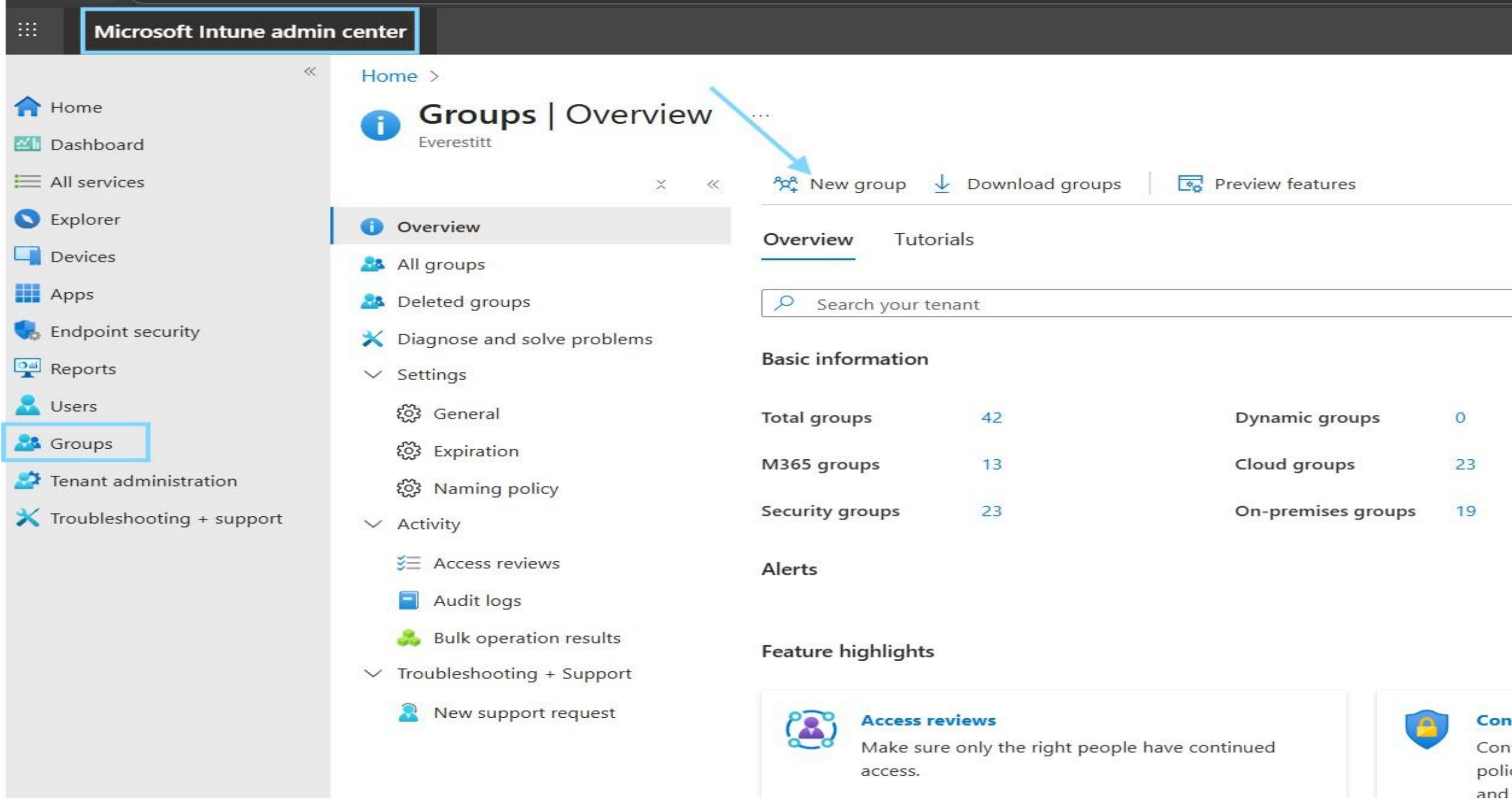
Group-Based Policy Targeting

Creating dedicated groups isolates test devices, streamlining policy application and reducing configuration errors.

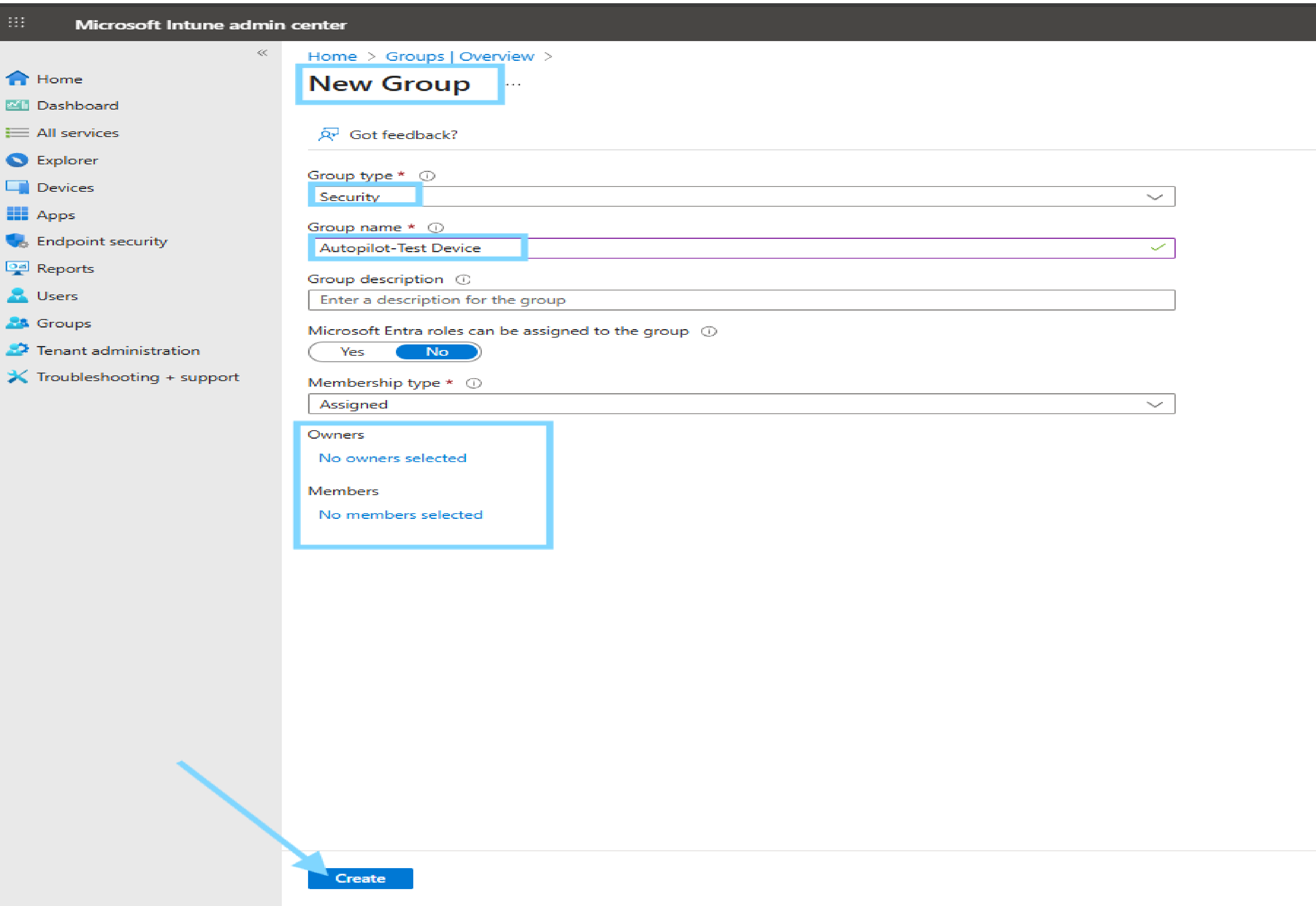
Minimizing Administrative Overhead

Group membership controls ensure only intended devices receive provisioning profiles, lowering administrative workload.

Create a group named Autopilot-Test Devices. After enrolling devices in Intune, organizing them into groups helps you manage and apply policies more effectively.



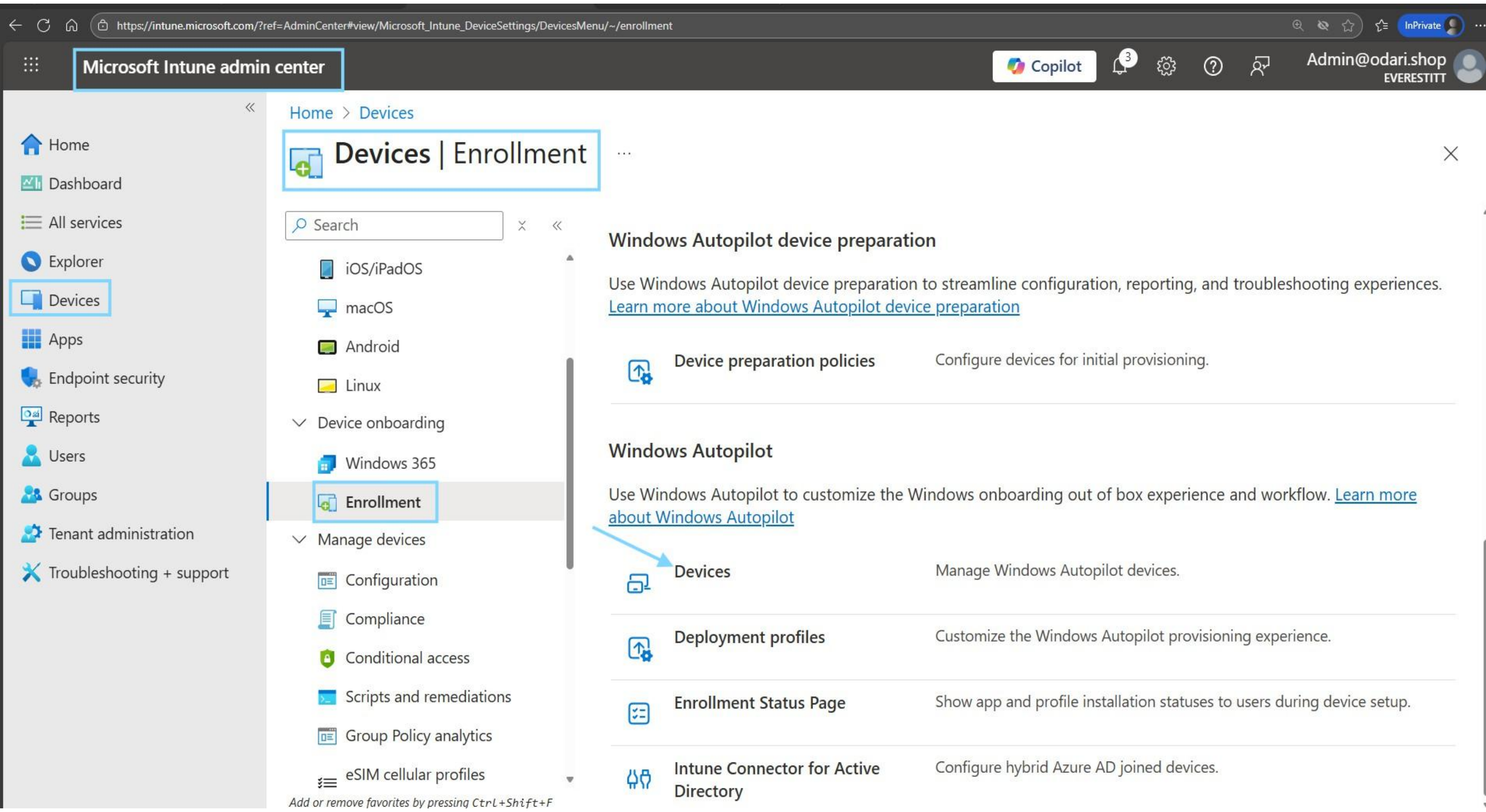
A security group named Autopilot-Test Devices is created to assign Autopilot profiles, ESP, and applications.



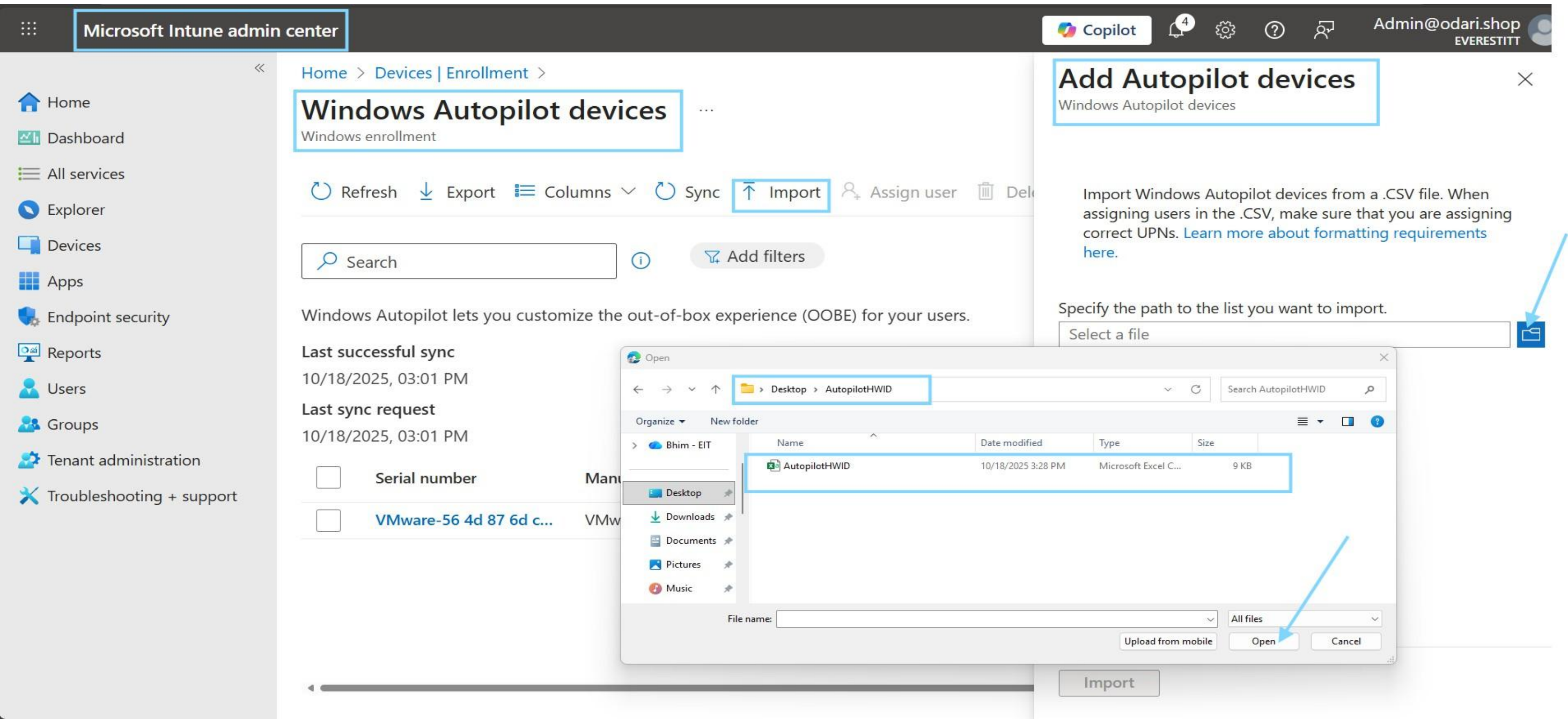
Go to Intune Admin Center: https://intune.microsoft.com

- Navigate to: Devices > Windows > Windows enrollment > Devices
  Click Import and upload your HWID .csv file

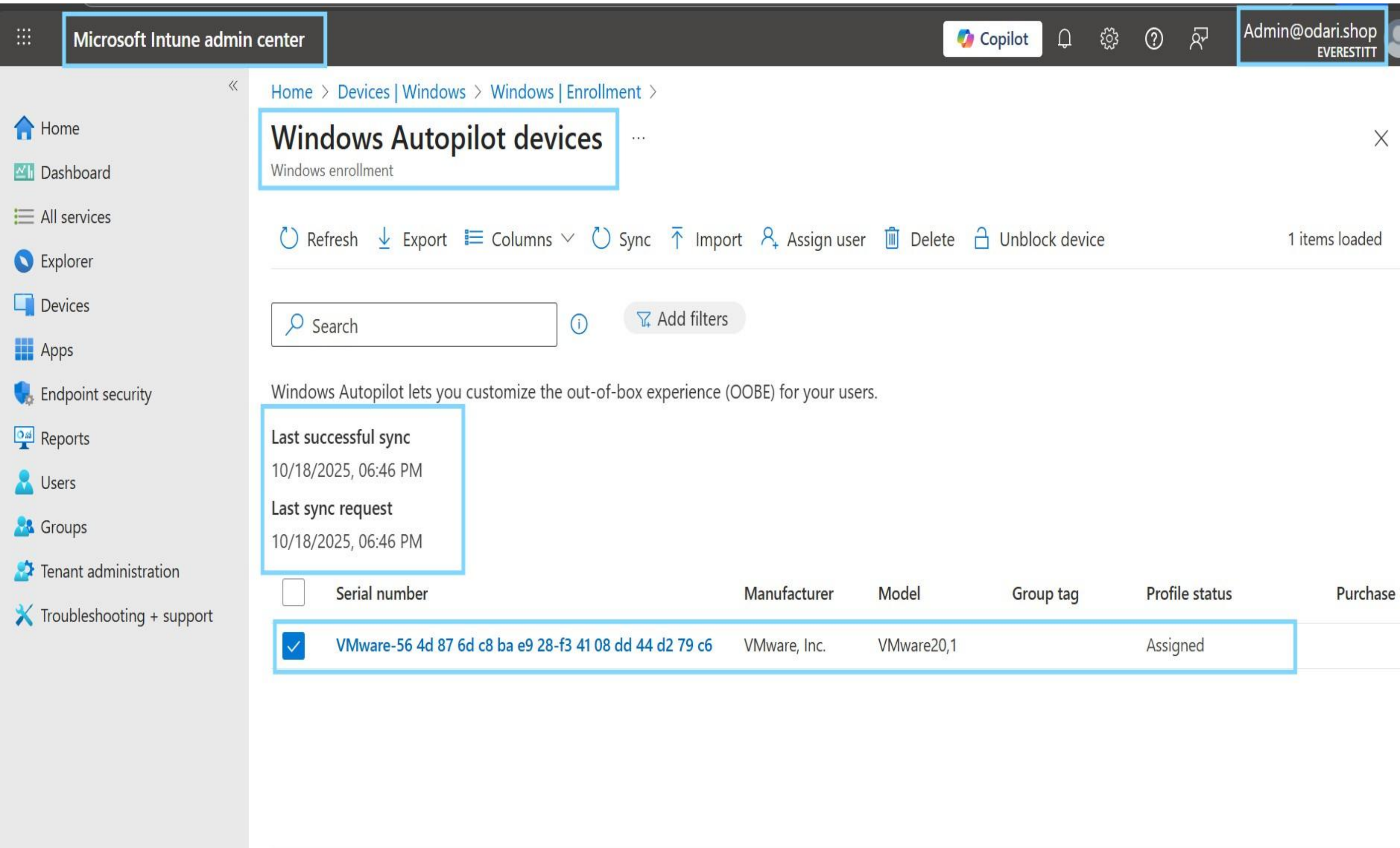
- Wait for processing, then assign an Autopilot profile to the device



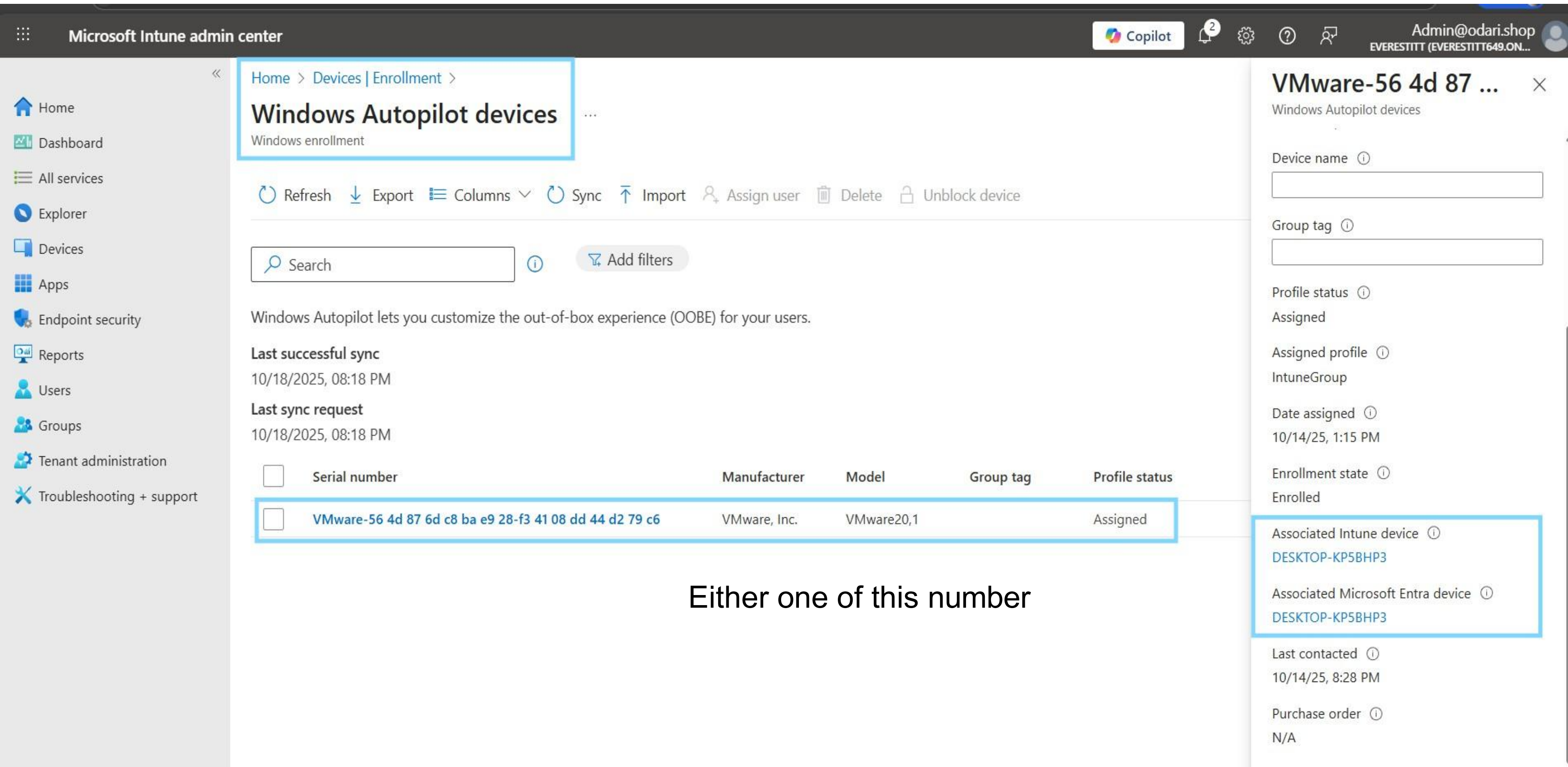
Importing the hardware hash registers the device in Windows Autopilot, enabling automated enrollment and provisioning through Intune.



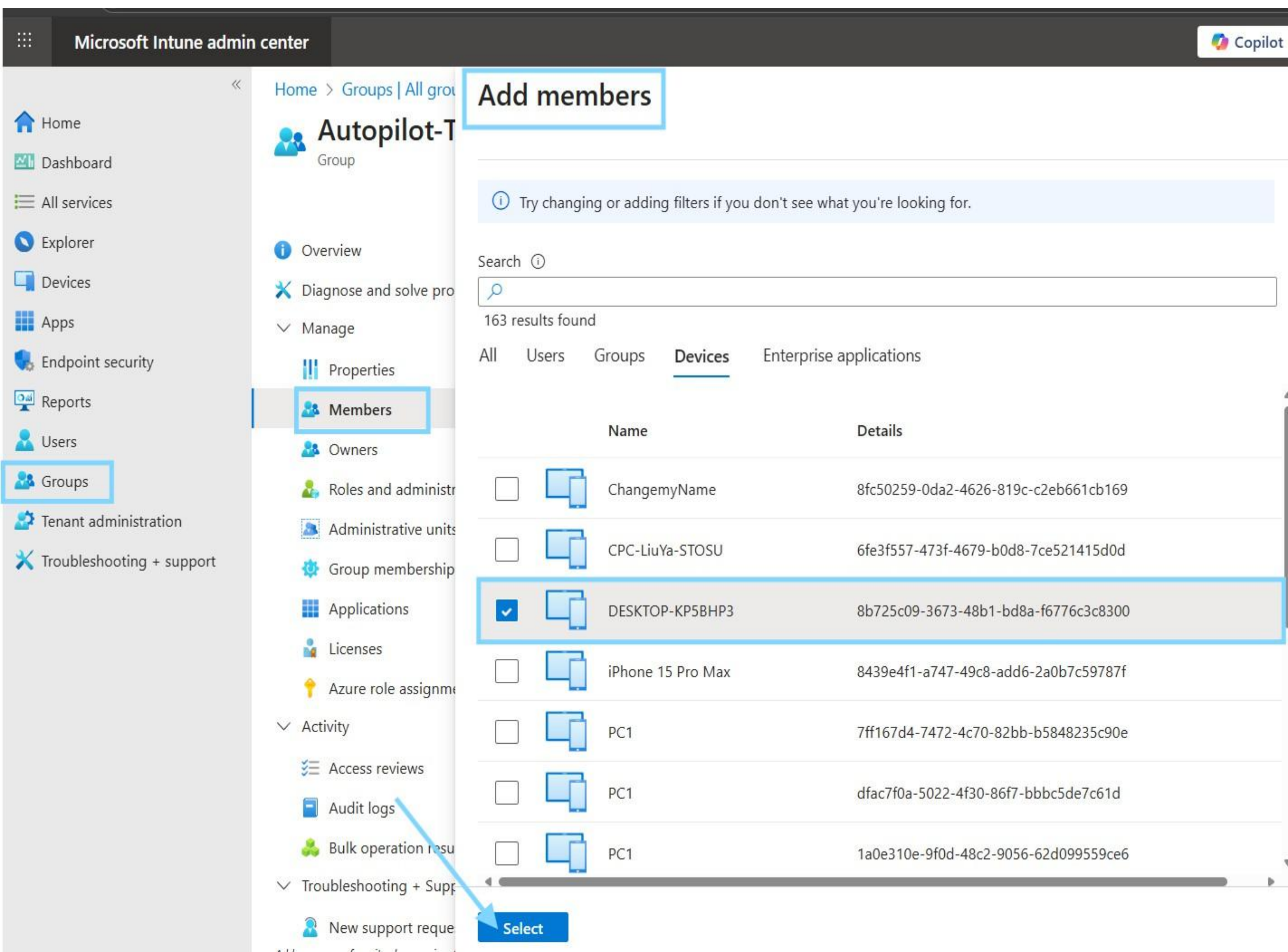
The device is successfully registered in Windows Autopilot and ready for automated deployment through Intune.



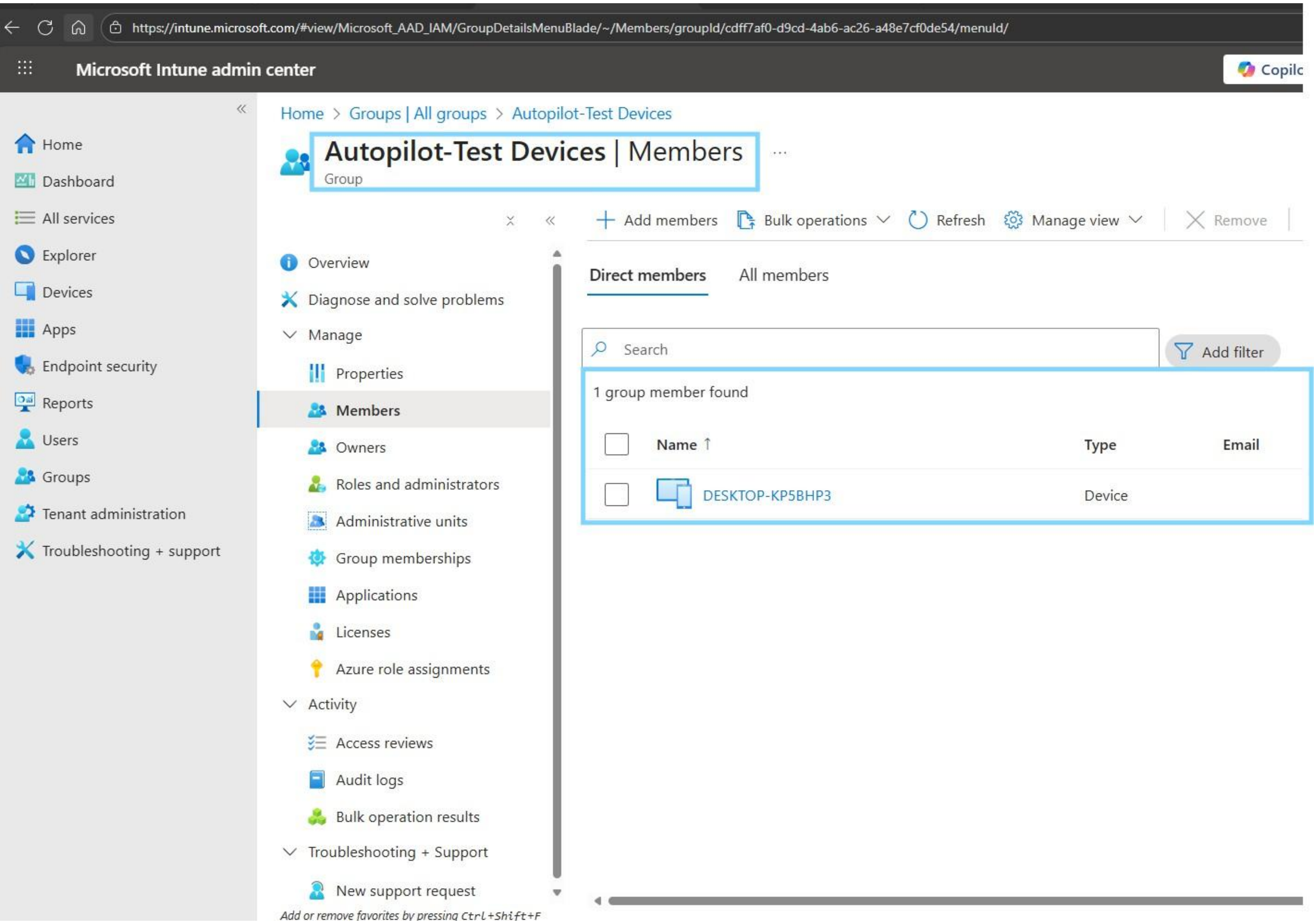
The HWID import was successful. The VMware virtual machine is now listed under Windows Autopilot devices in Intune, showing its serial number, manufacturer (VMware, Inc.), and model.

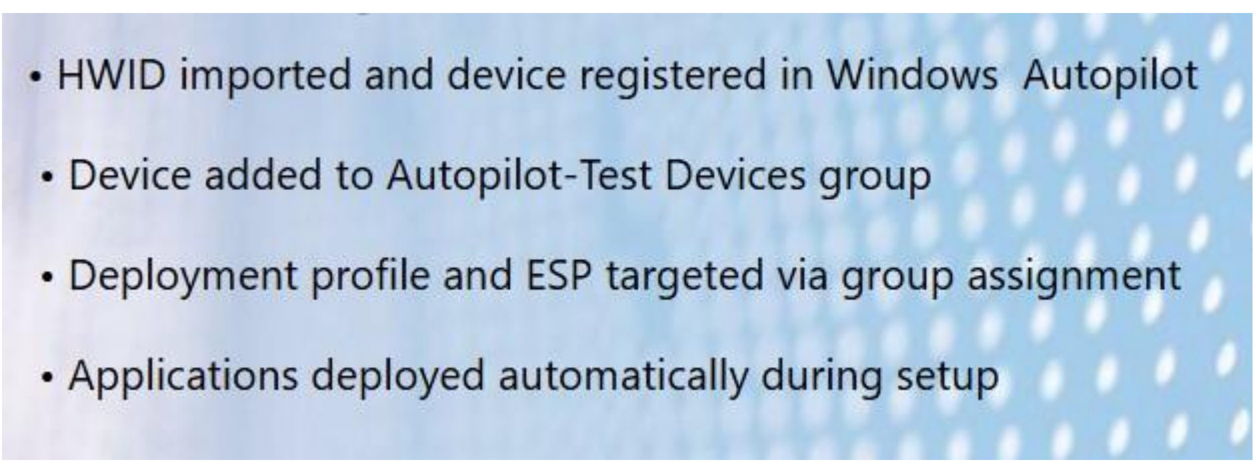


The next step is to add members to the appropriate group to ensure the right users or devices are included in the deployment and receive assigned policies, apps, and configurations.

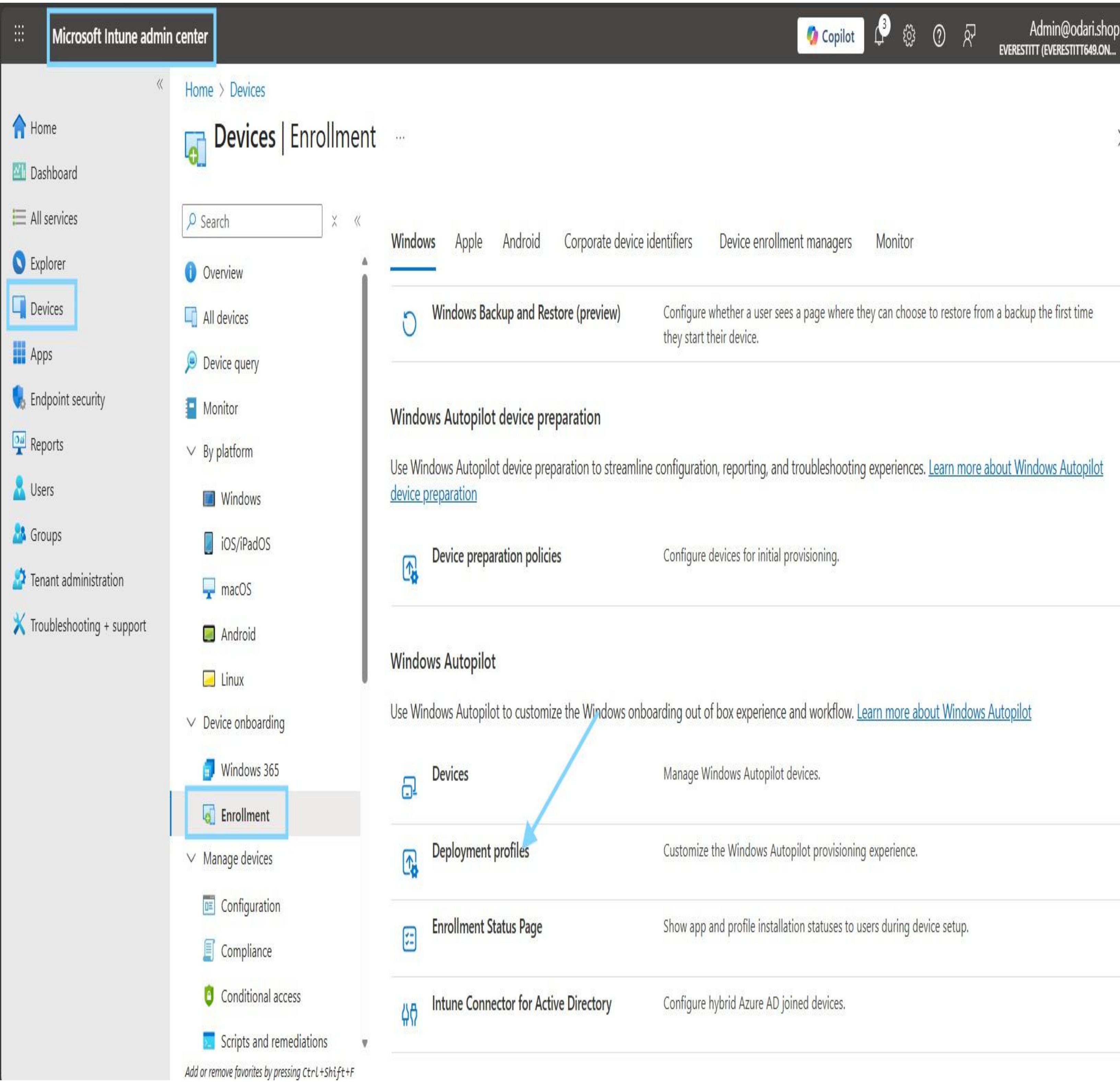


This Intune device group (Autopilot-Test Devices) with a Windows device added as a direct member for Windows Autopilot testing and management.

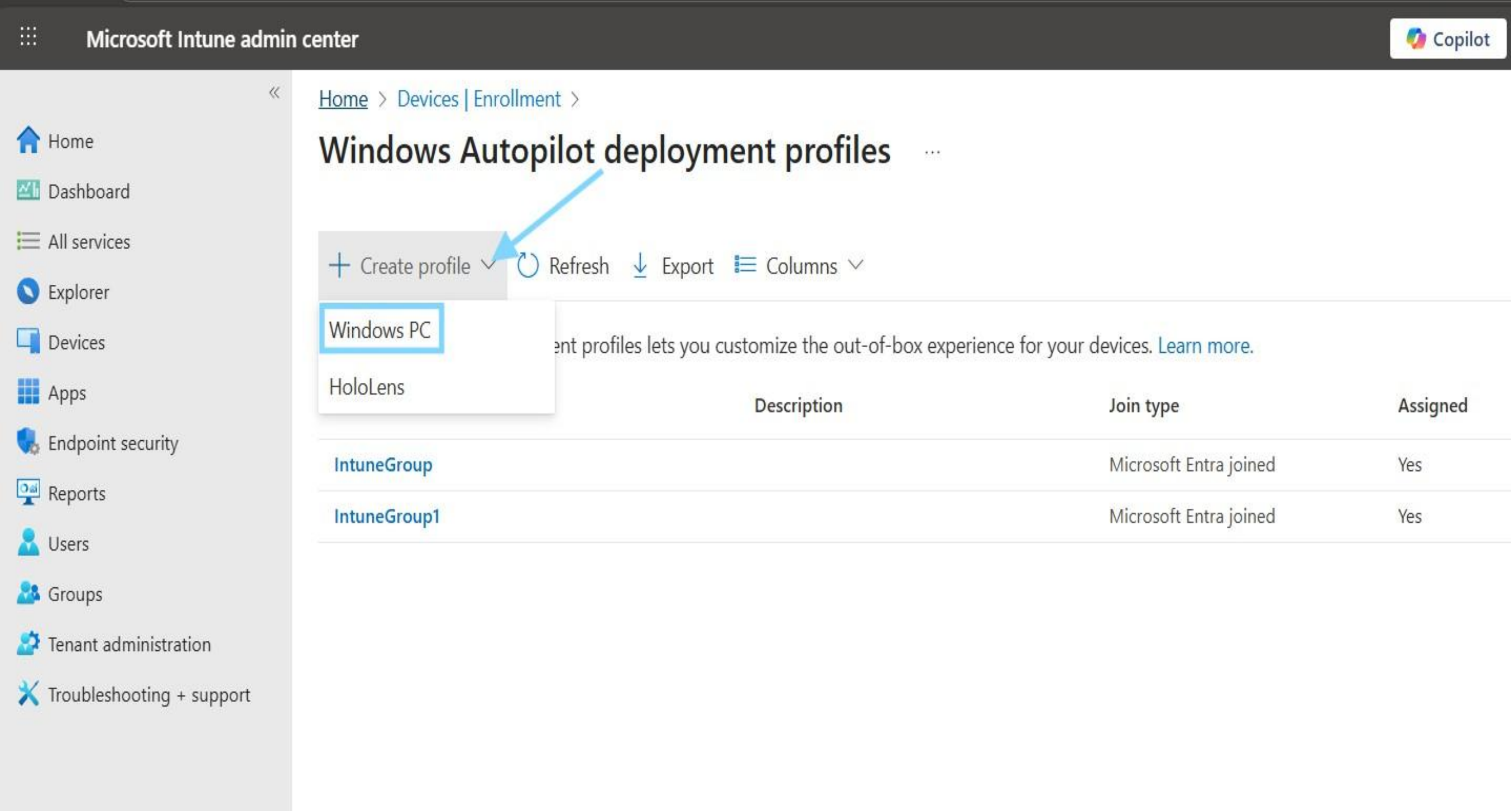




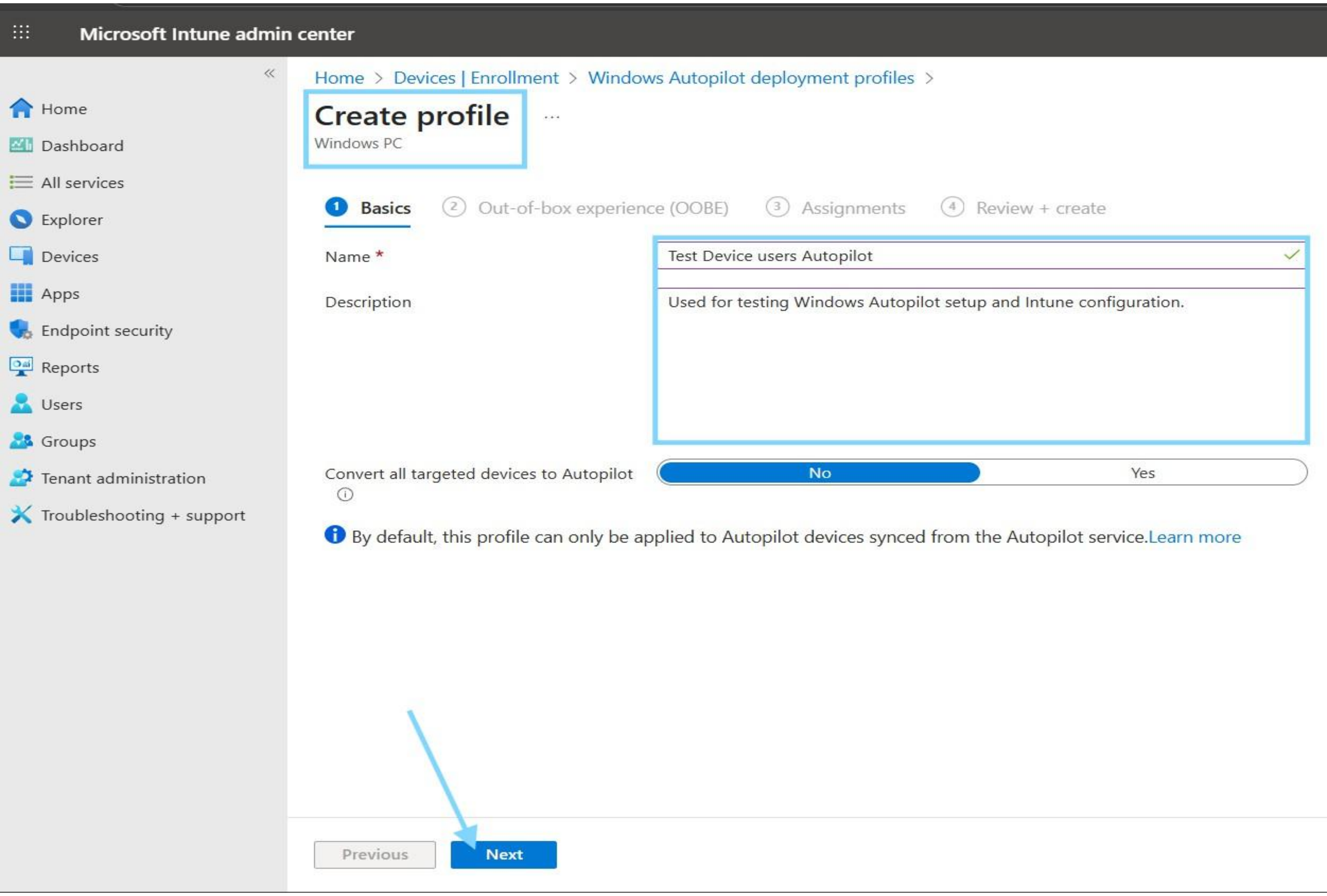
This configures and manages Windows Autopilot enrollment settings, including deployment profiles and device onboarding in Microsoft Intune.

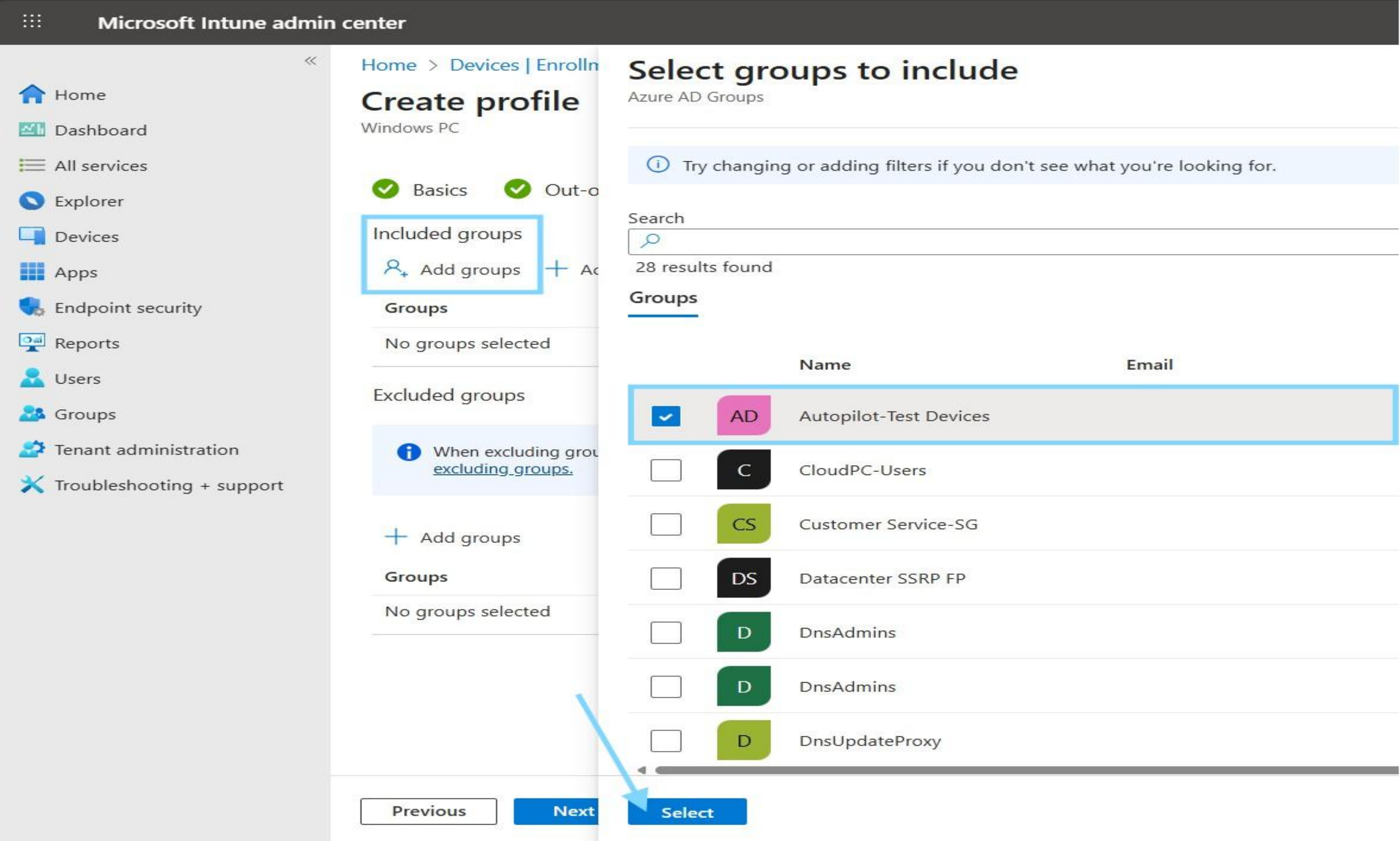


To create and manage Windows Autopilot deployment profiles to define the out-of-box setup and Entra ID join experience for Windows devices.

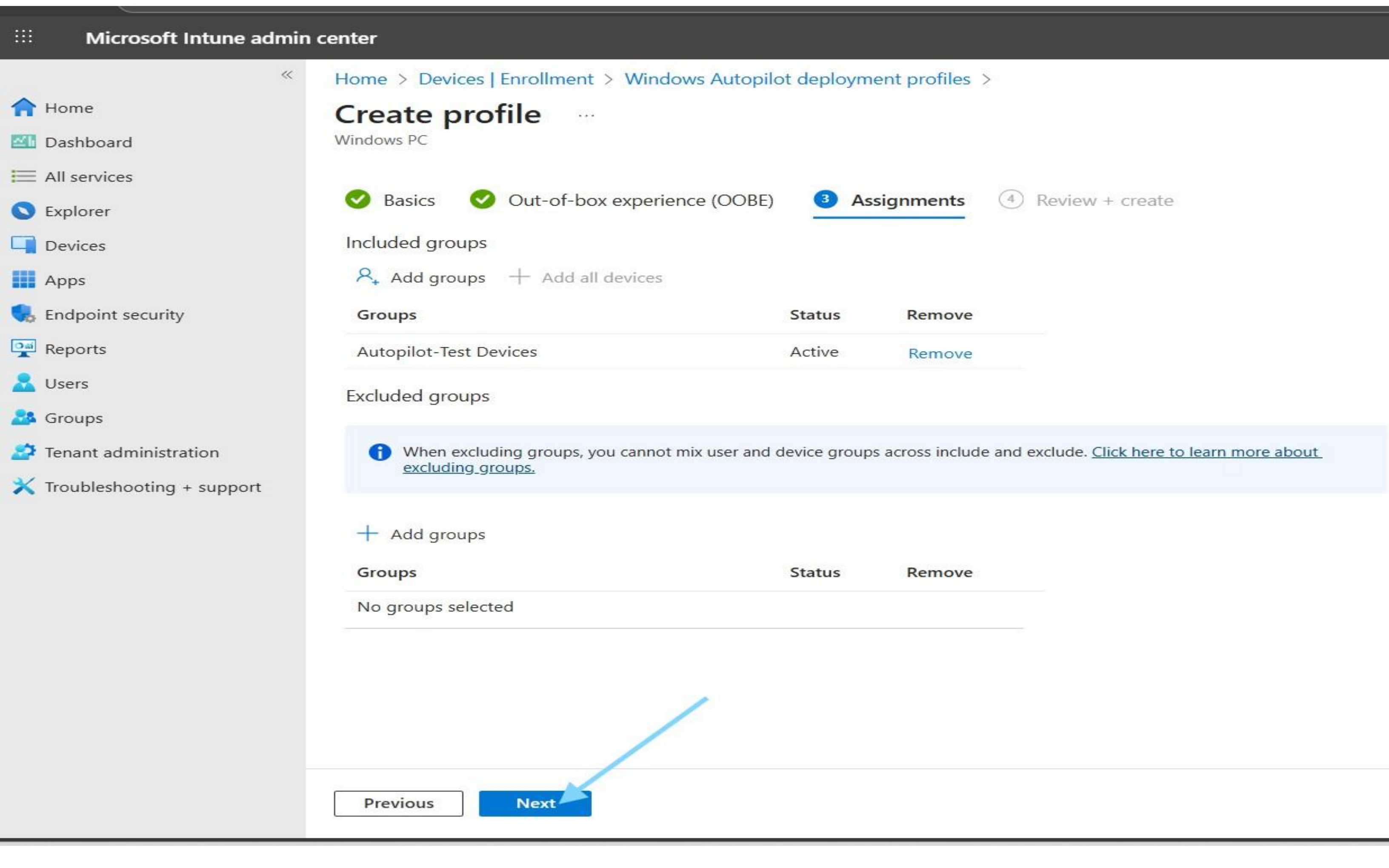


To create a Windows Autopilot deployment profile by defining basic details for testing device enrollment and Intune configuration before setup begins.

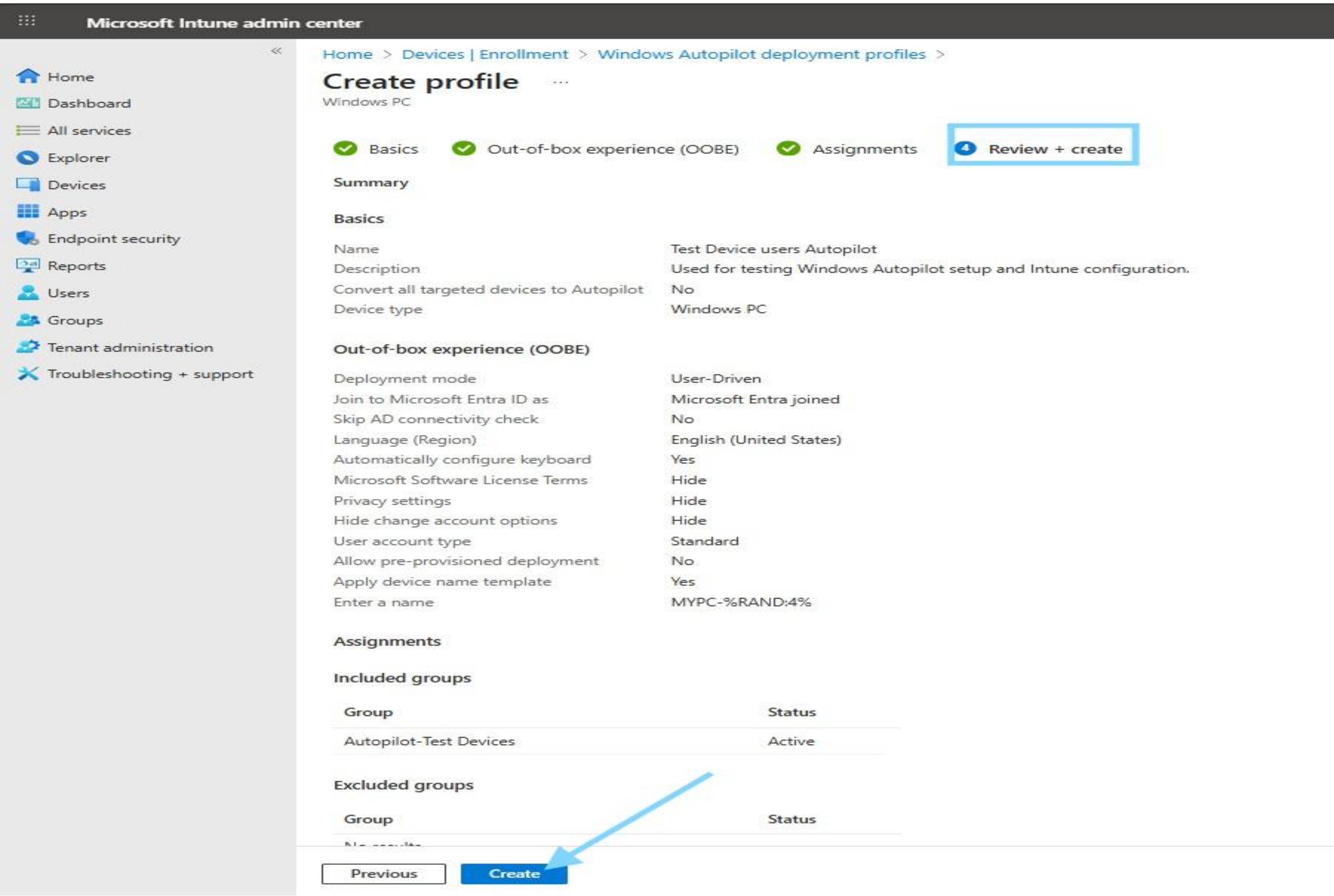




This assigns the Autopilot profile to the Autopilot-Test Devices group so only those devices receive the deployment settings.



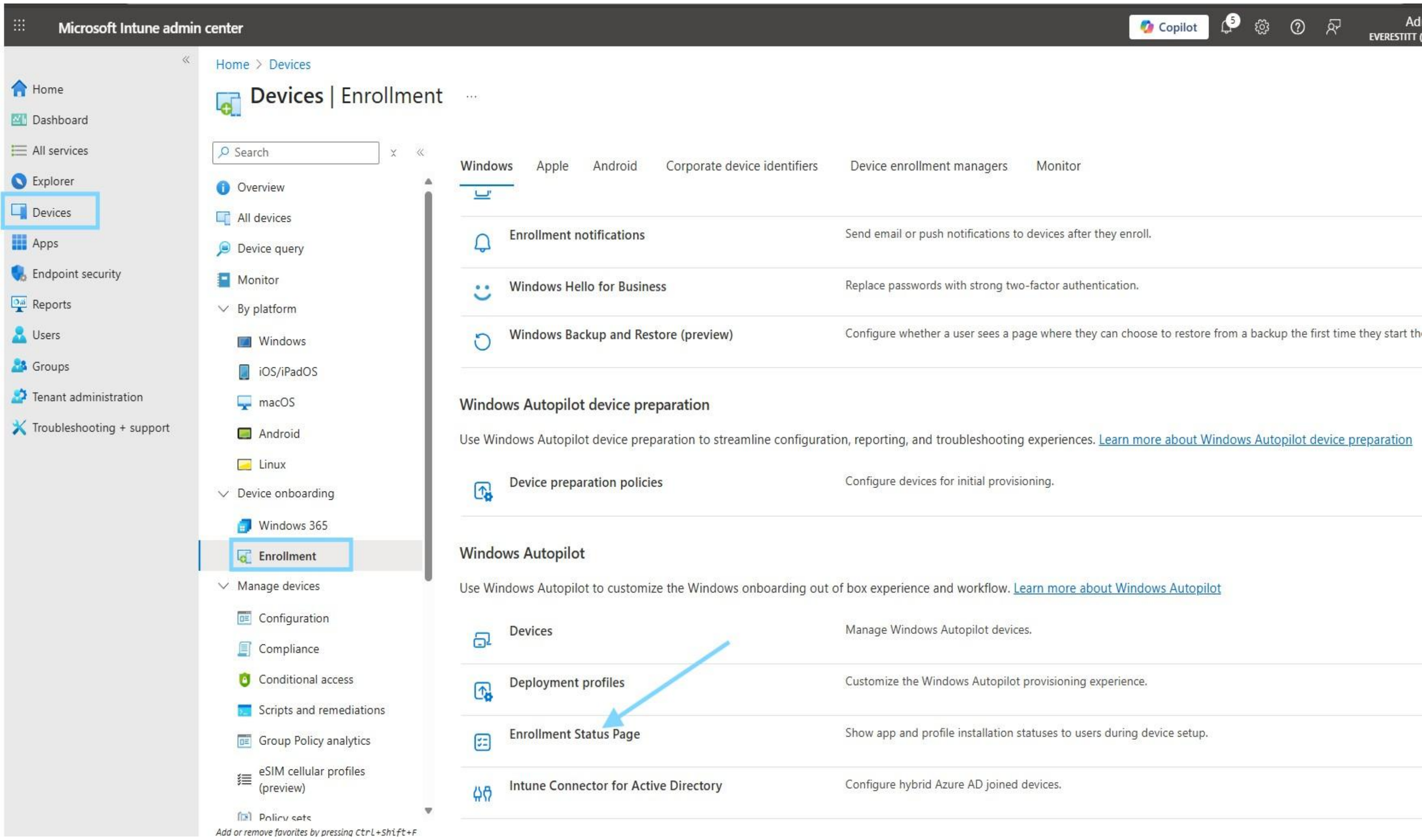
This is the final review screen where you confirm all Autopilot profile settings and click Create to finish creating the deployment profile.



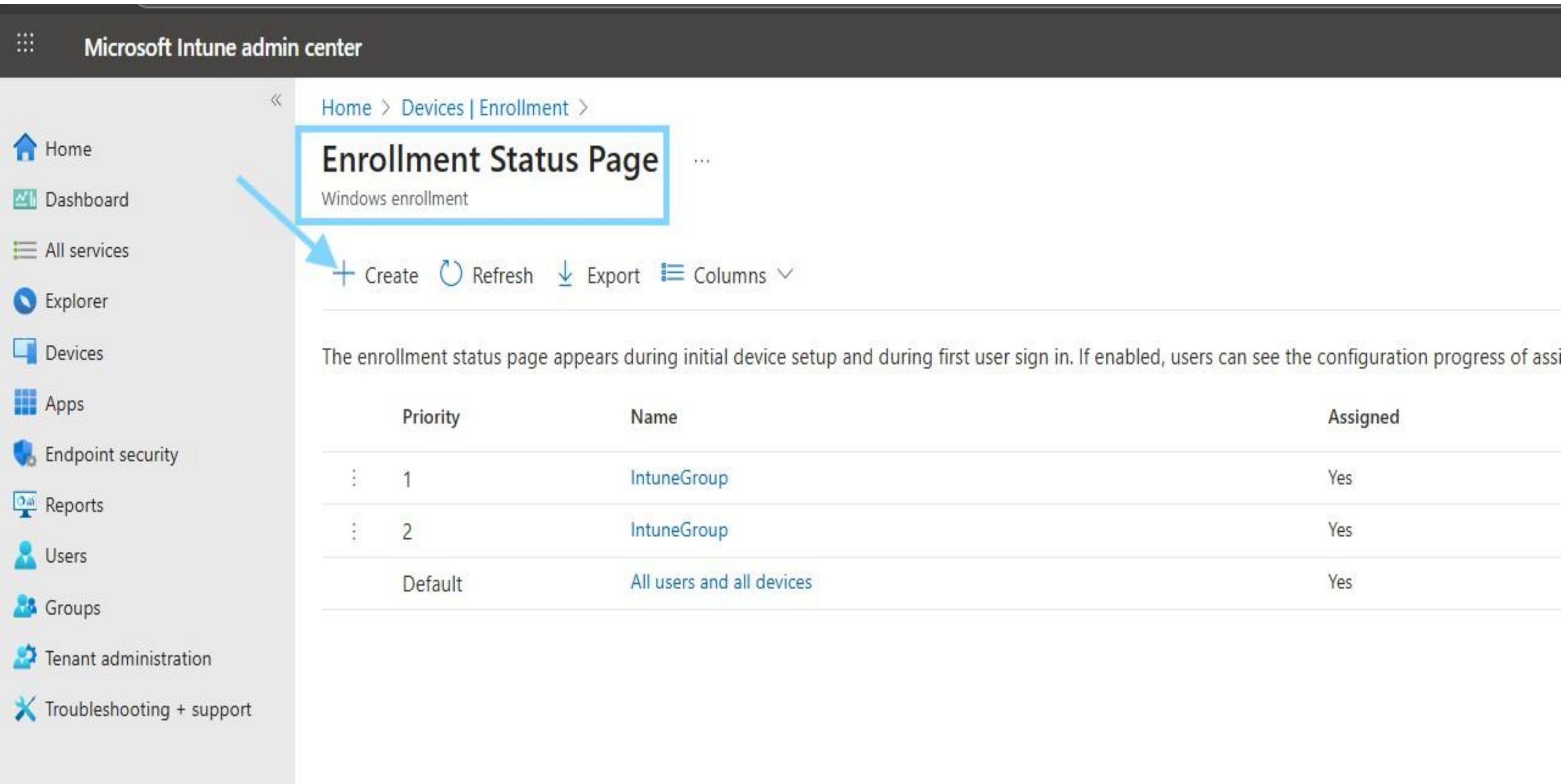
Profile creation is complete your Autopilot deployment profile is ready and assigned to the Autopilot-Test Devices group.

The Enrollment Status Page (**ESP**) in Intune is used to control what happens during the Windows Autopilot setup process after a device is enrolled. It ensures that required apps, policies, and configurations are installed before the user can access the desktop. This helps guarantee compliance and readiness before the device is handed over for use.

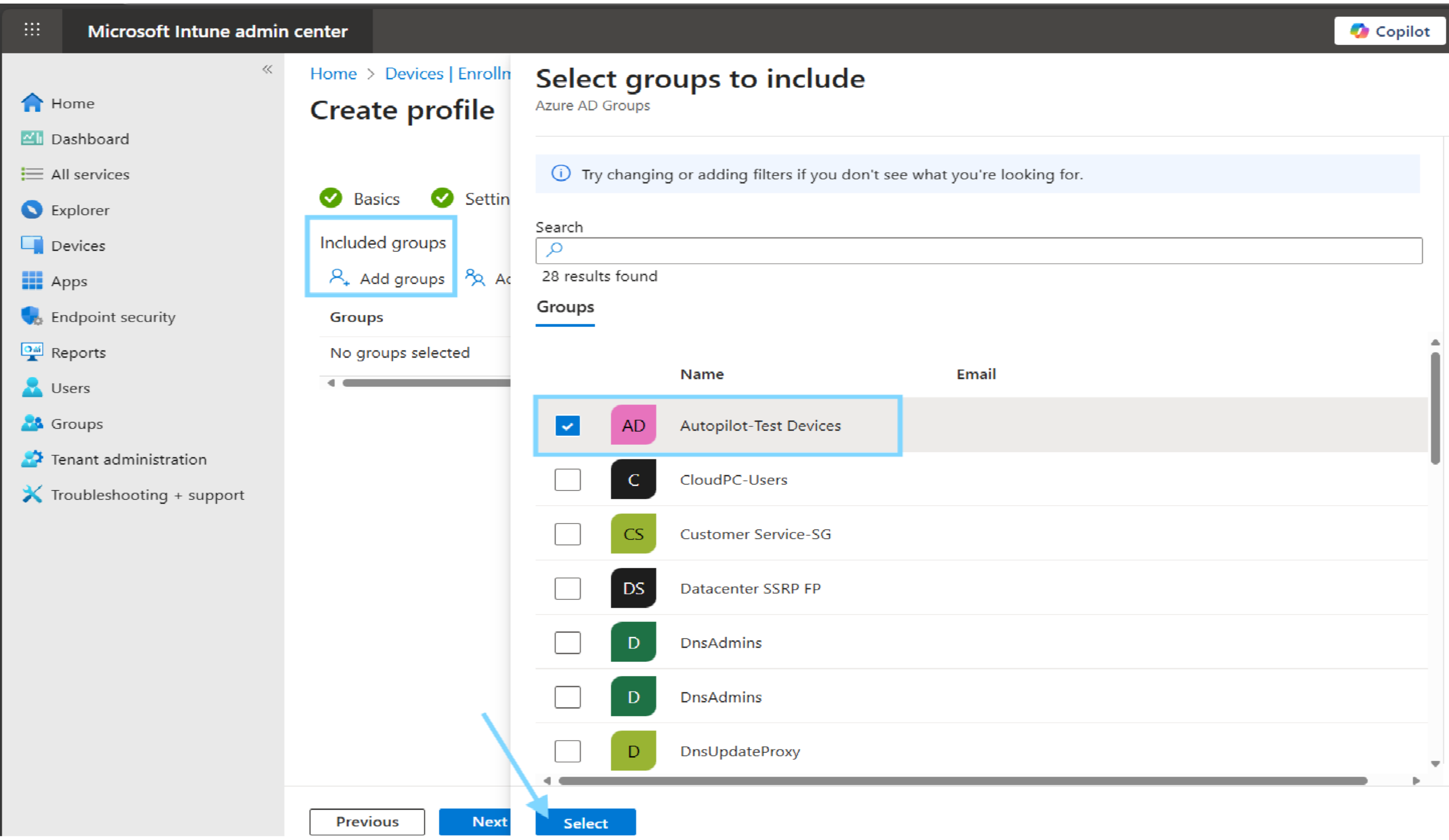
Windows Autopilot enrollment section in Microsoft Intune, where administrators manage Autopilot devices, deployment profiles, and the Enrollment Status Page for automated Windows provisioning.

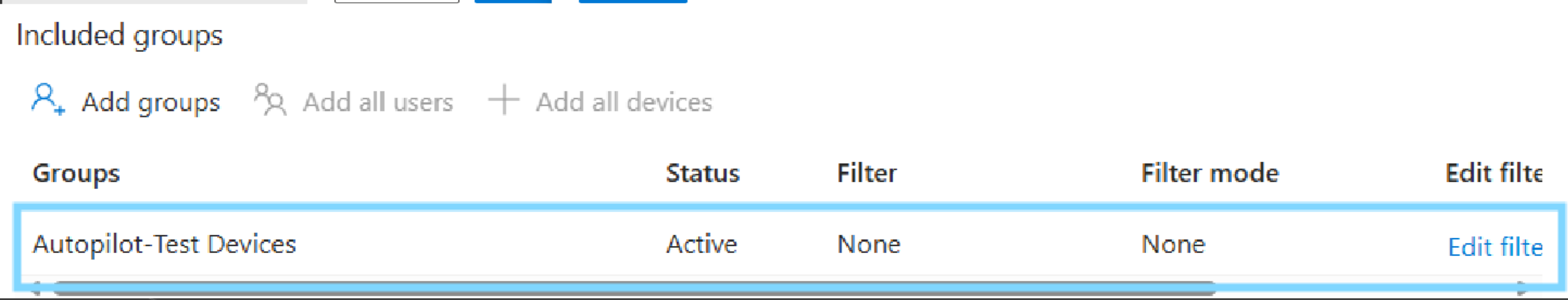


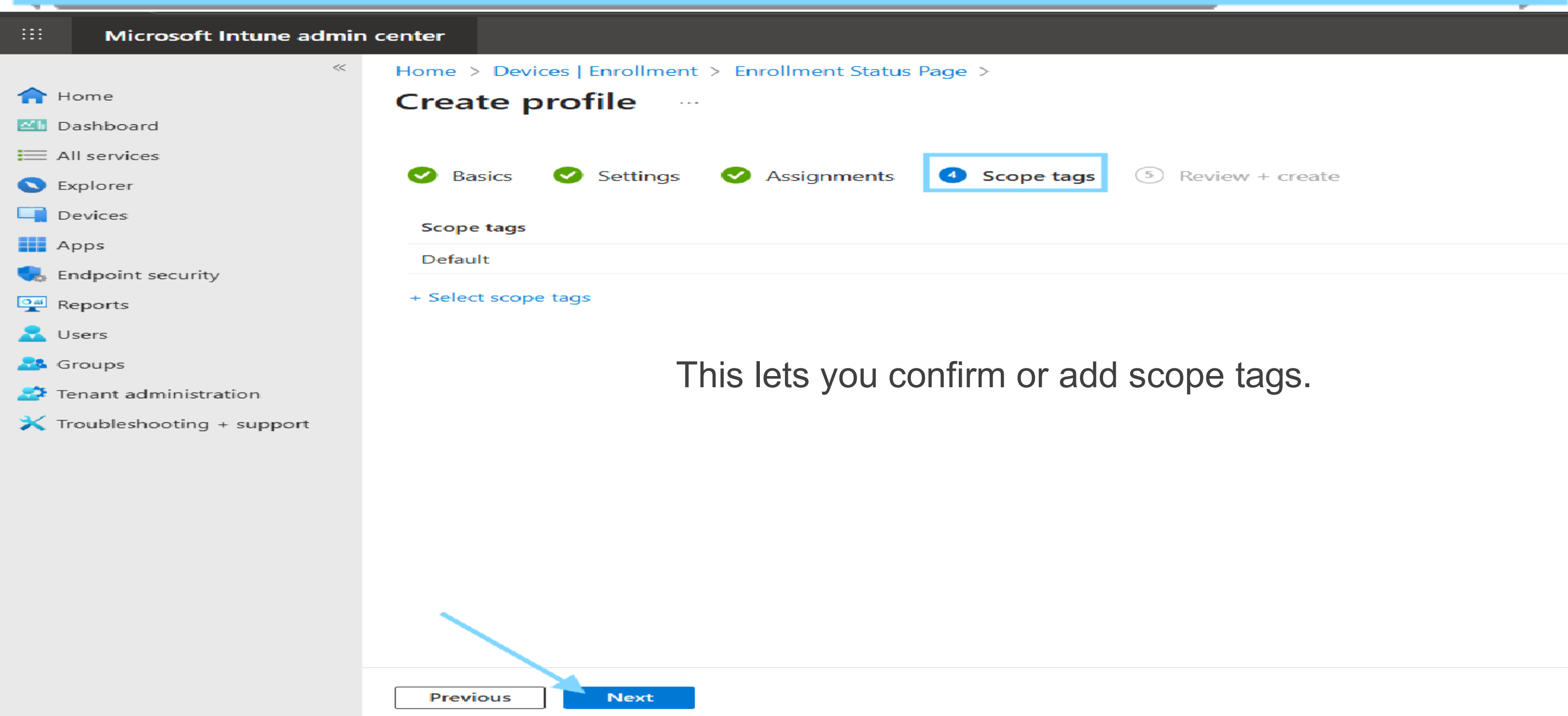
- Click Create to add a new profile for managing the setup experience.



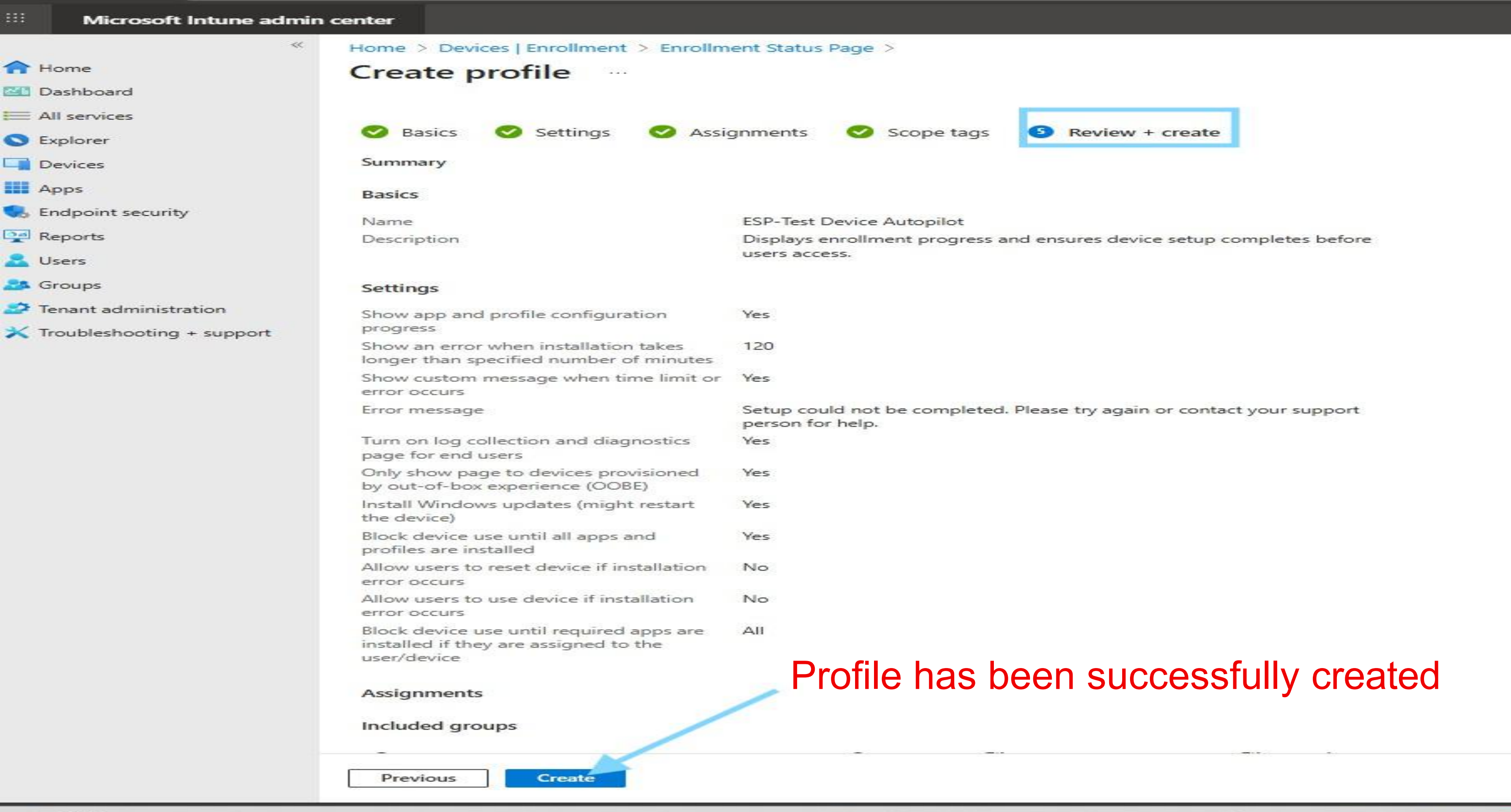
This selects the Autopilot-Test Devices group, so the Autopilot profile applies only to those devices.



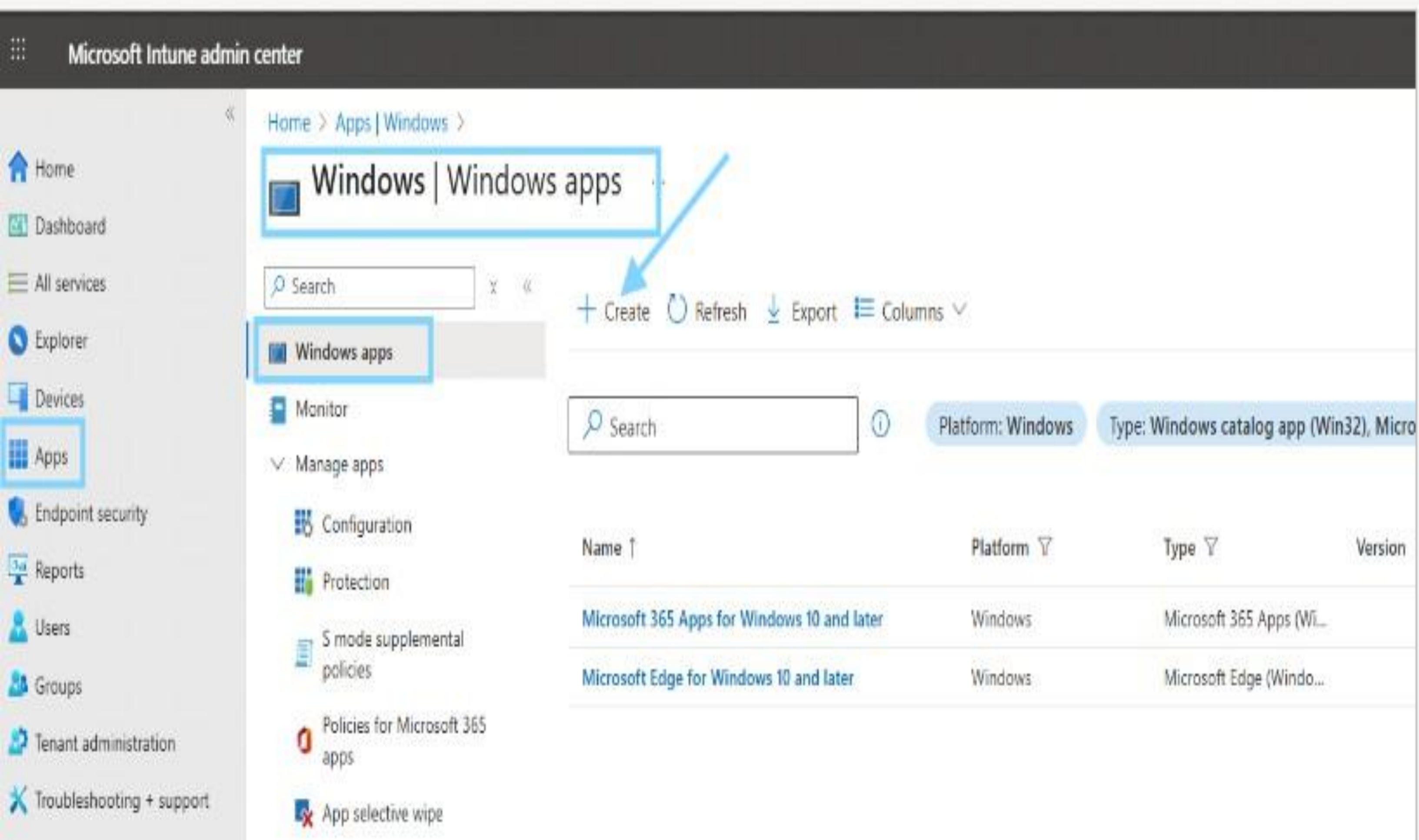


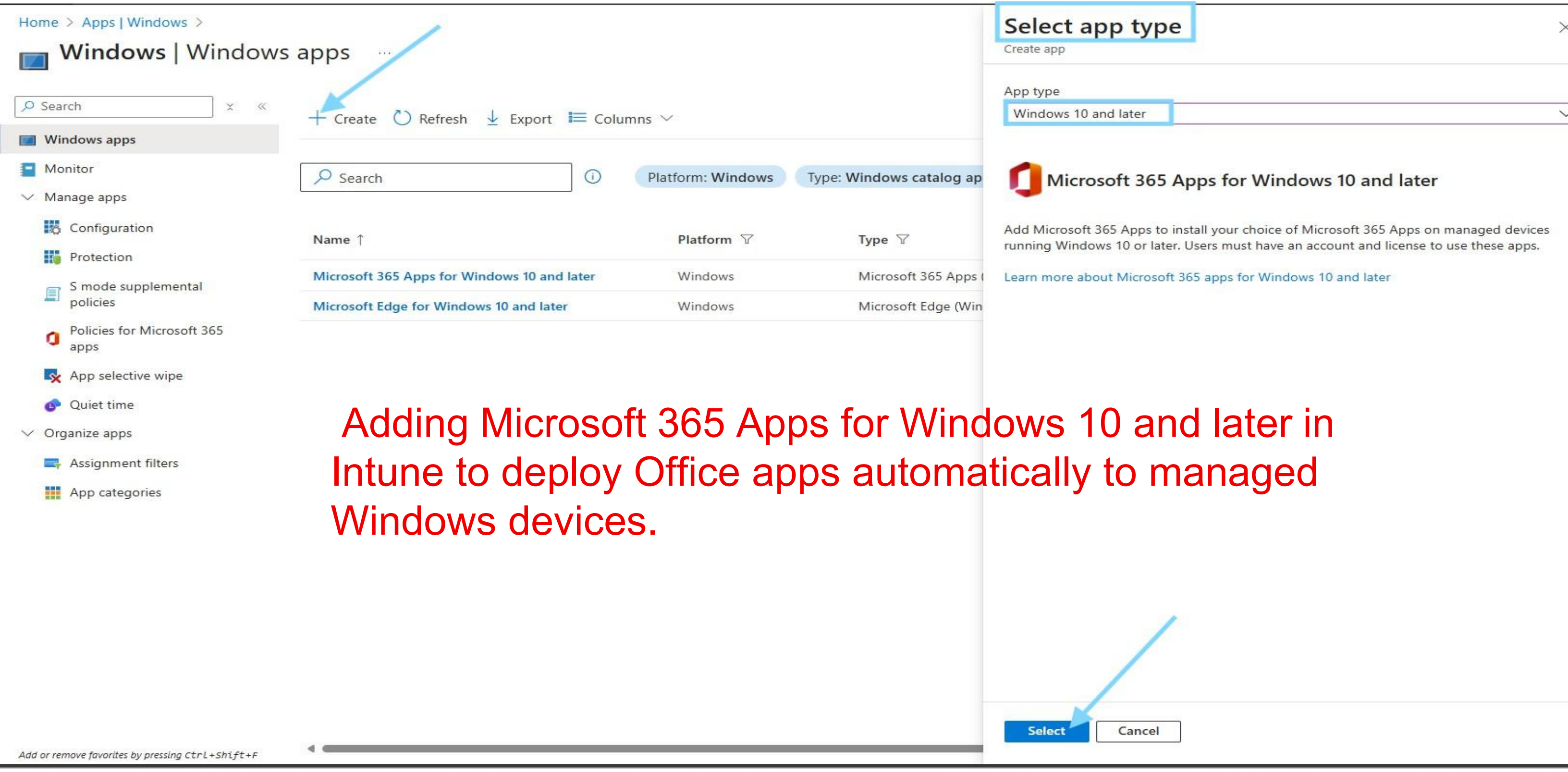


This lets you confirm or add scope tags.



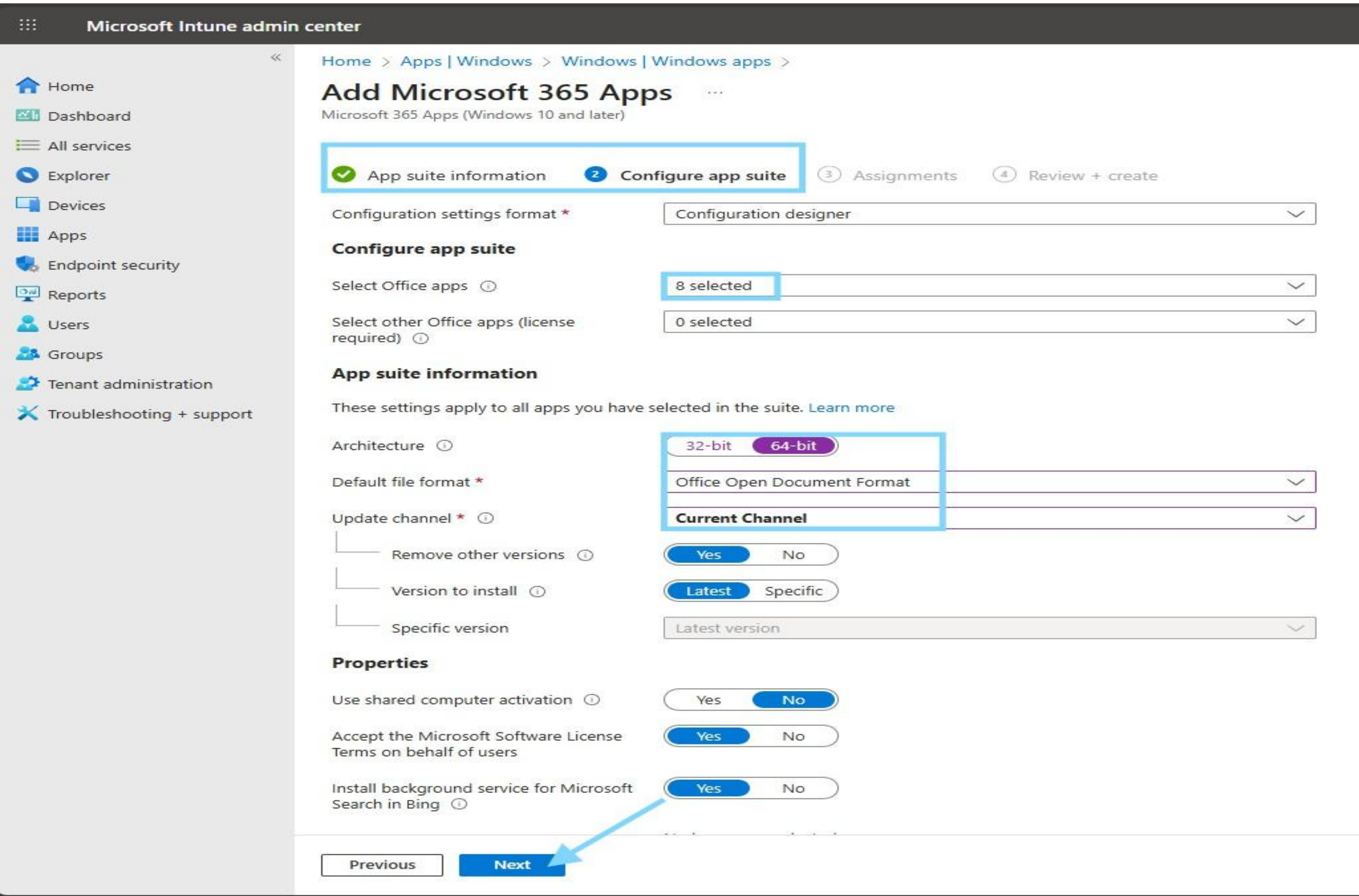
To deploy an app using Intune, first go to the Intune Admin Center and find the app you want to install on users' devices. Make sure the app is set up correctly and ready for deployment. Tthen check that the installation settings match your organization's policies. Finally, choose the specific group of users or devices that should receive the app, so it only installs on those selected endpoints.



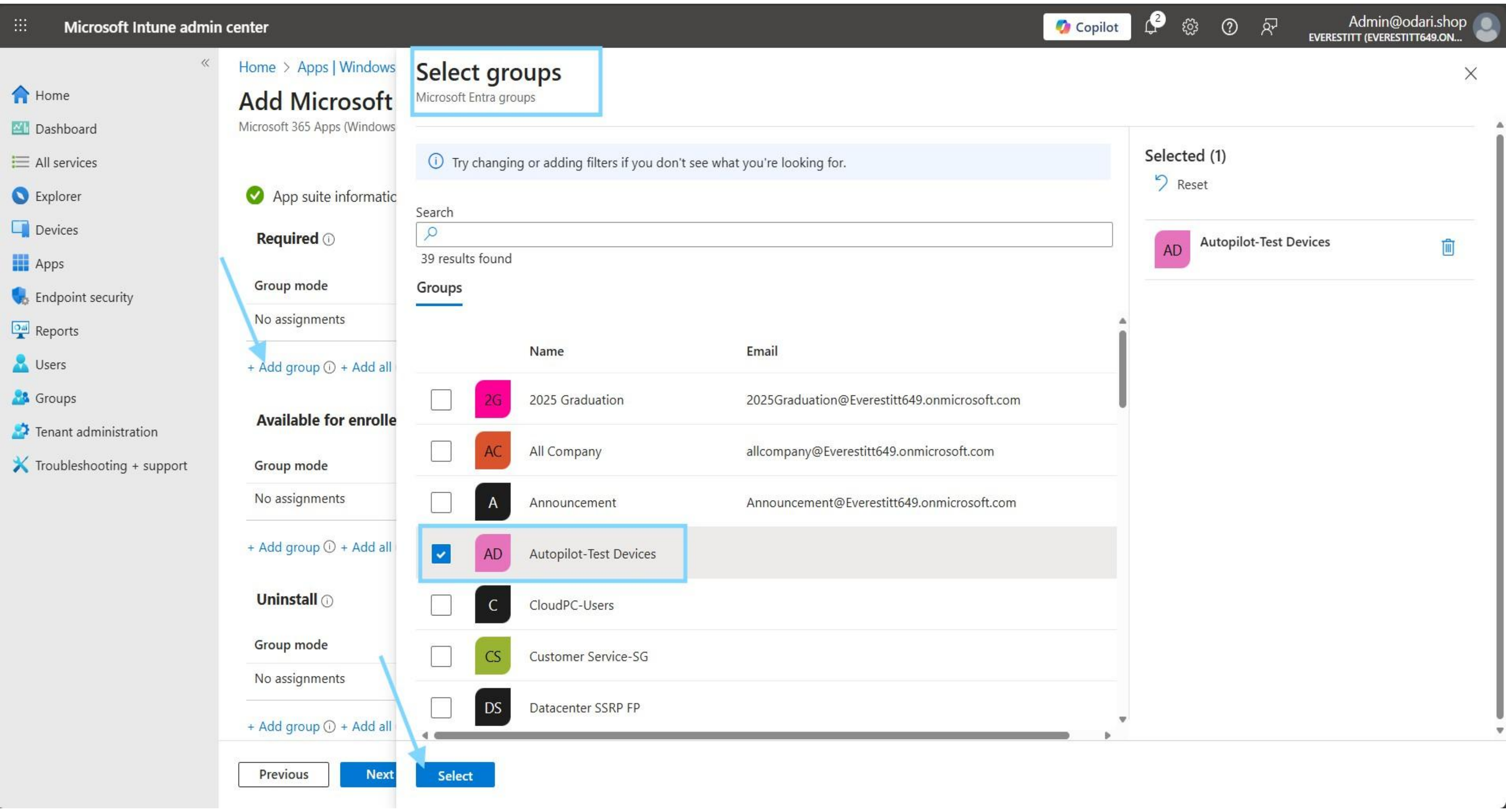


Adding Microsoft 365 Apps for Windows 10 and later in Intune to deploy Office apps automatically to managed Windows devices.

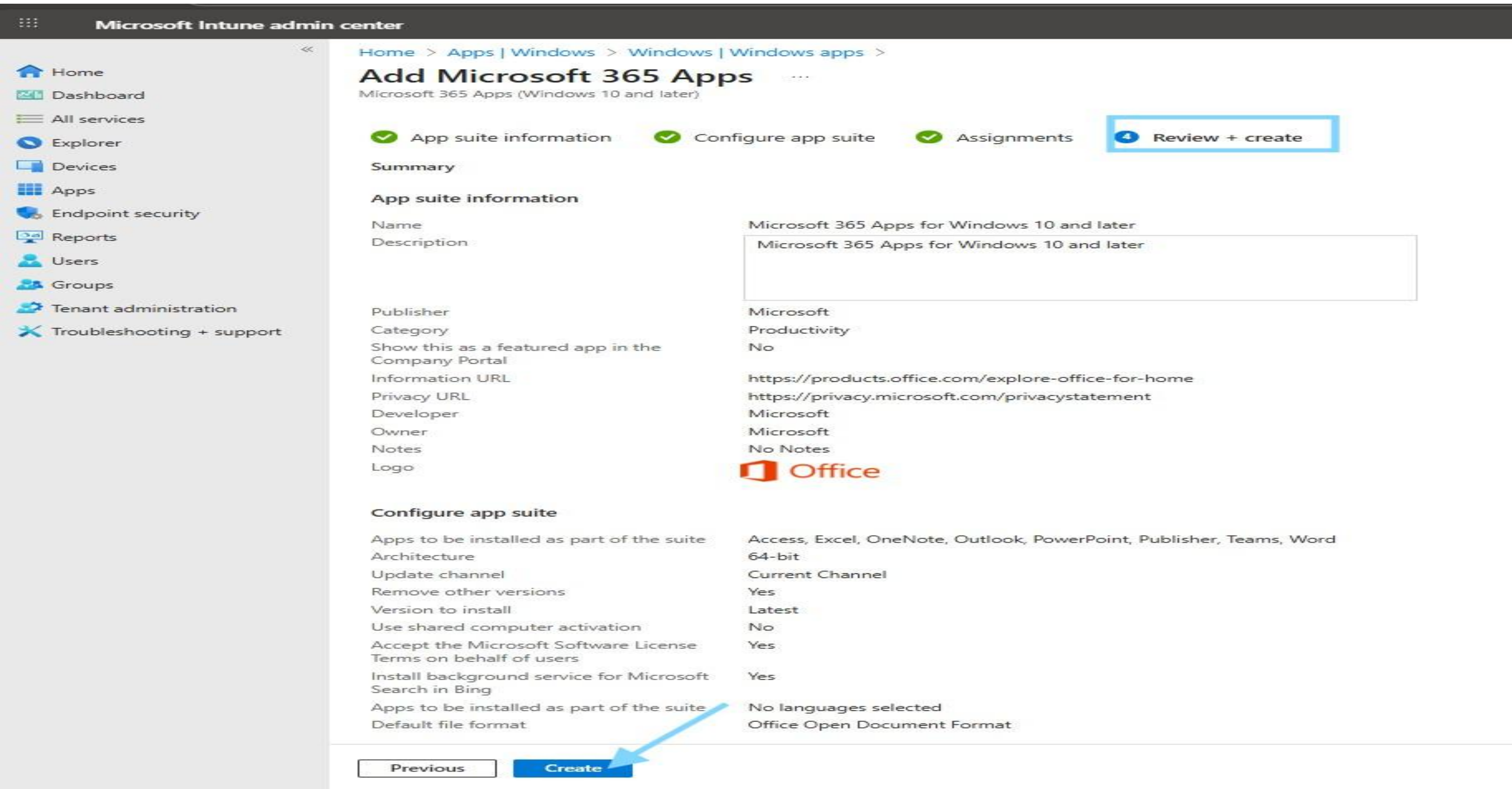
- Pick which Office apps to install, choose 64-bit architecture, set the update channel to Current Channel, and confirm options like removing old versions and accepting license terms. After these settings, you click Next to assign the apps to devices.



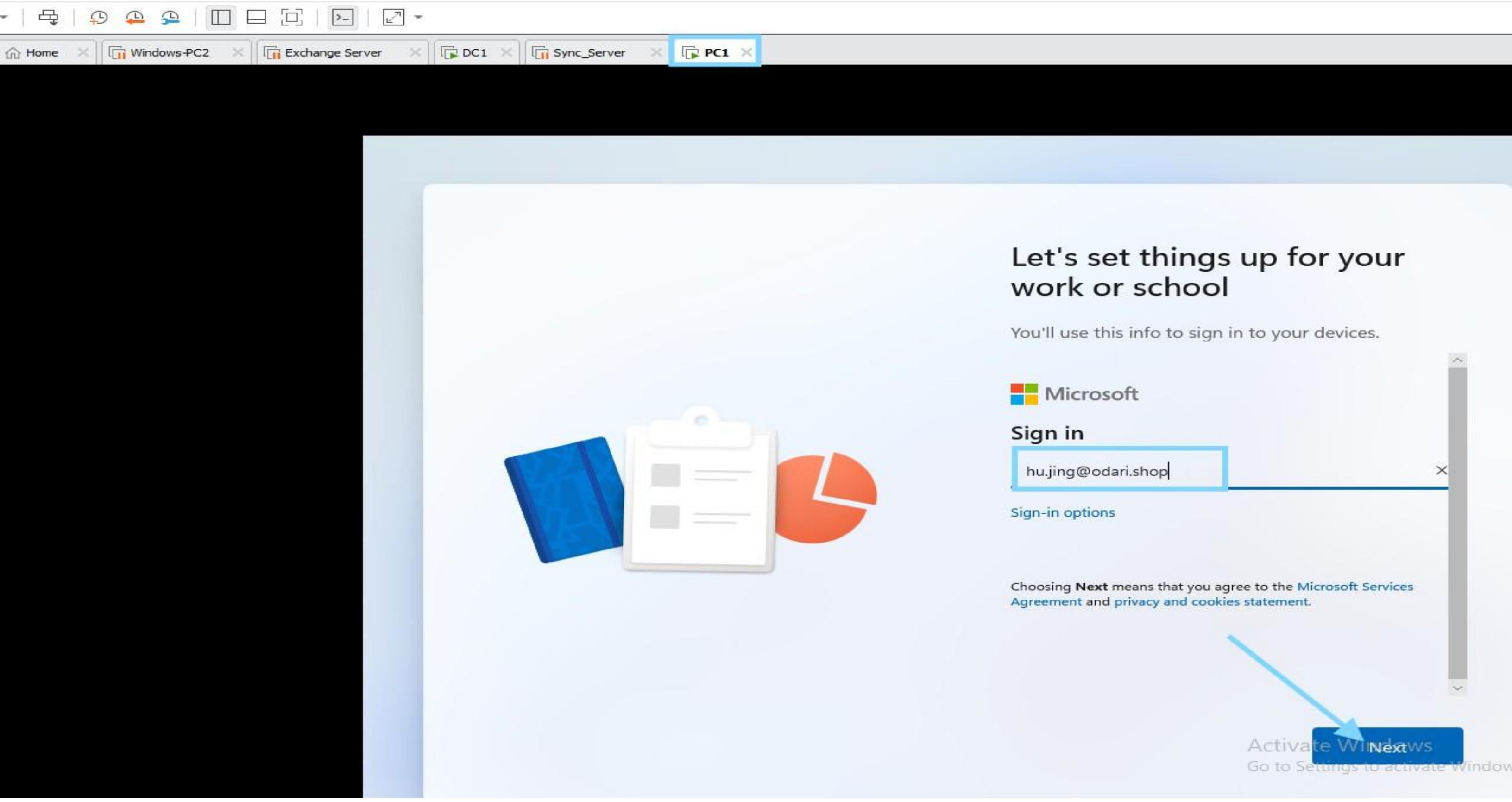
- Selecting the group Autopilot-Test Devices so that these devices will receive the Microsoft 365 Apps you configured earlier. After choosing the group, you click Select to confirm. This ensures only the devices in that group get the apps during deployment.



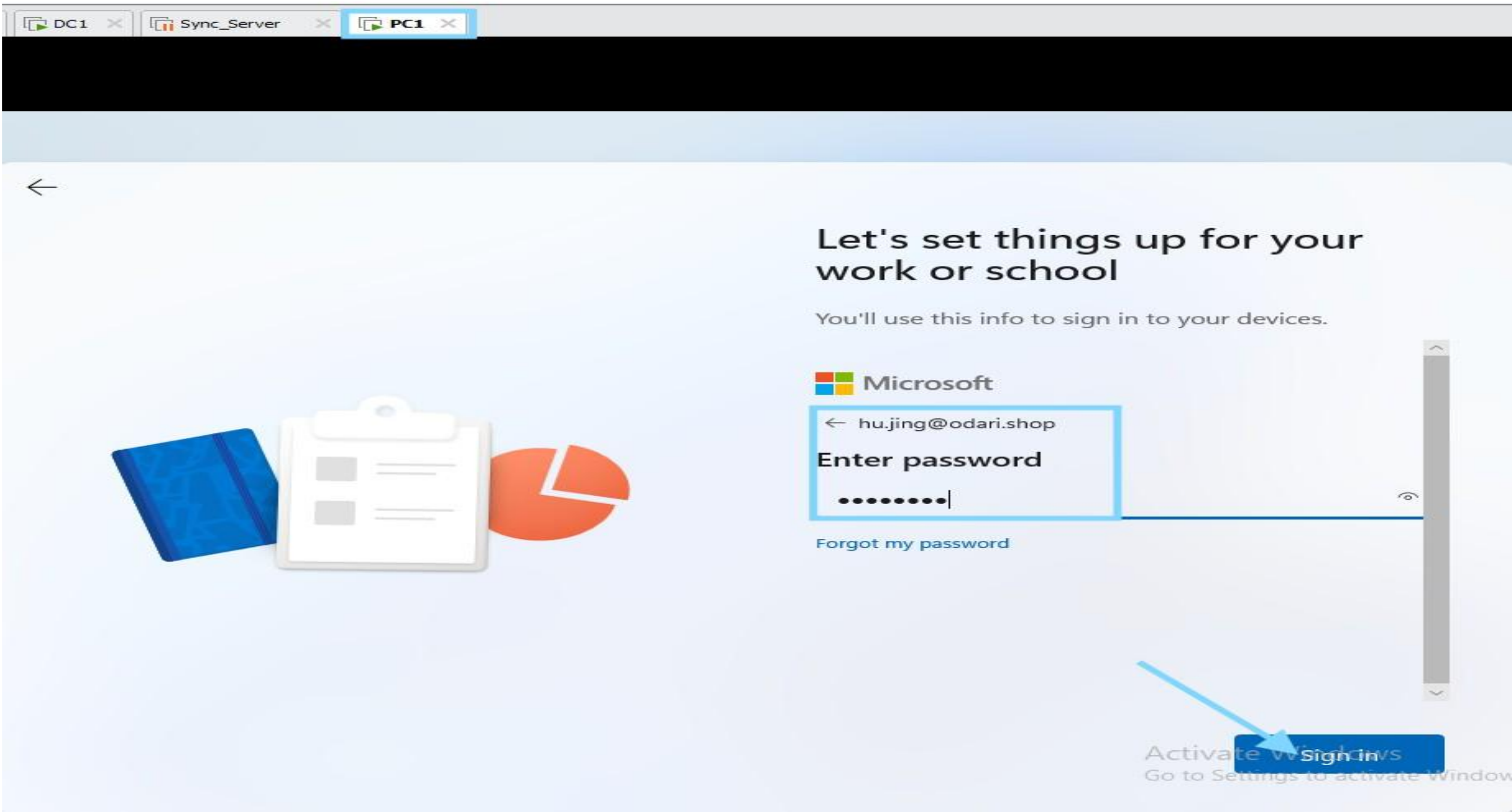
- Check all settings like selected Office apps, architecture (64-bit), update channel (Current Channel), and other options. If everything looks correct, click Create to finish and start deploying the apps.



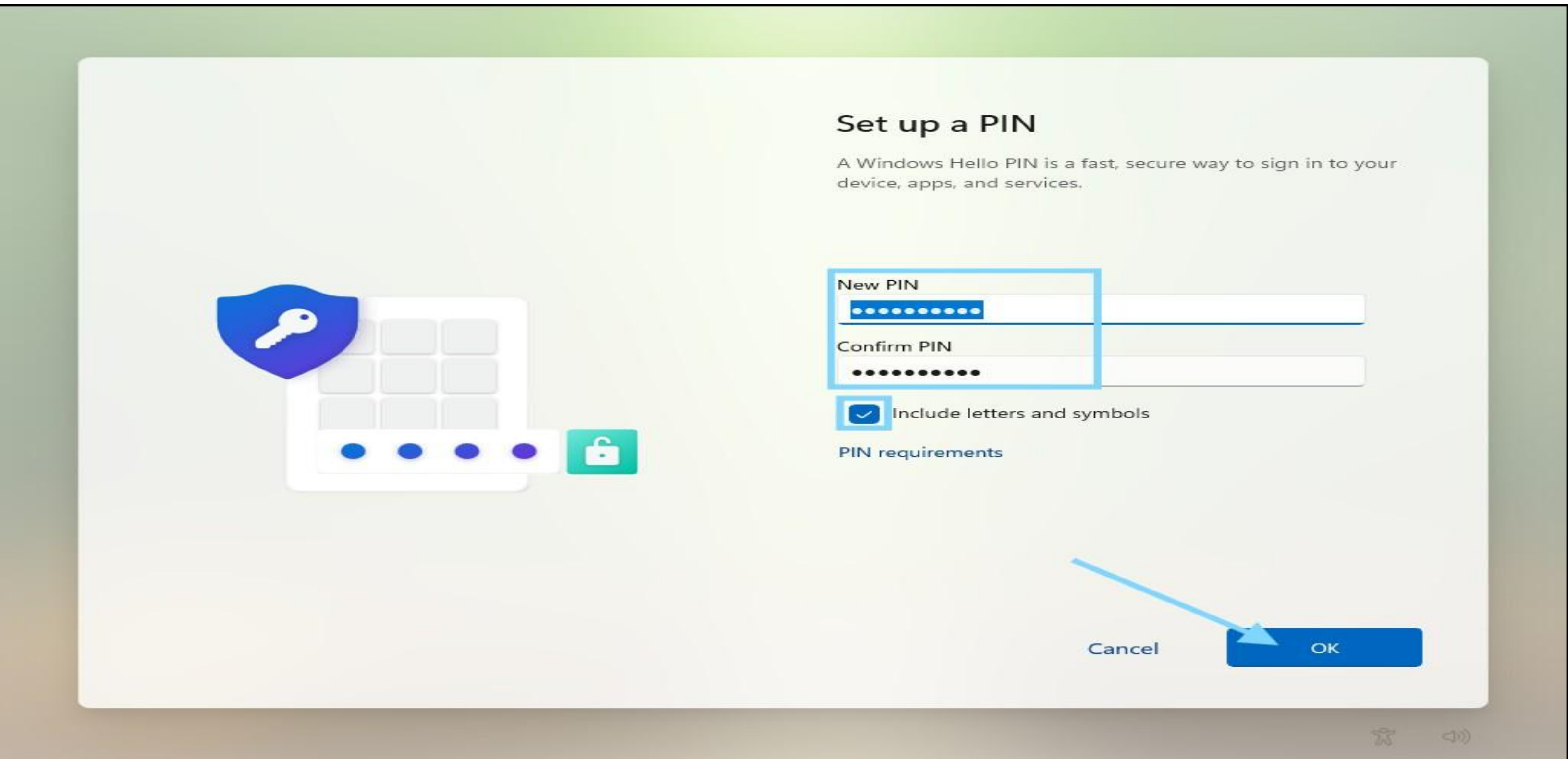
Hu Jing's computer has all the apps needed installed. When she logs in, she will be able to see them on her screen and start using them without any issues. Everything is set up and ready for her to work.



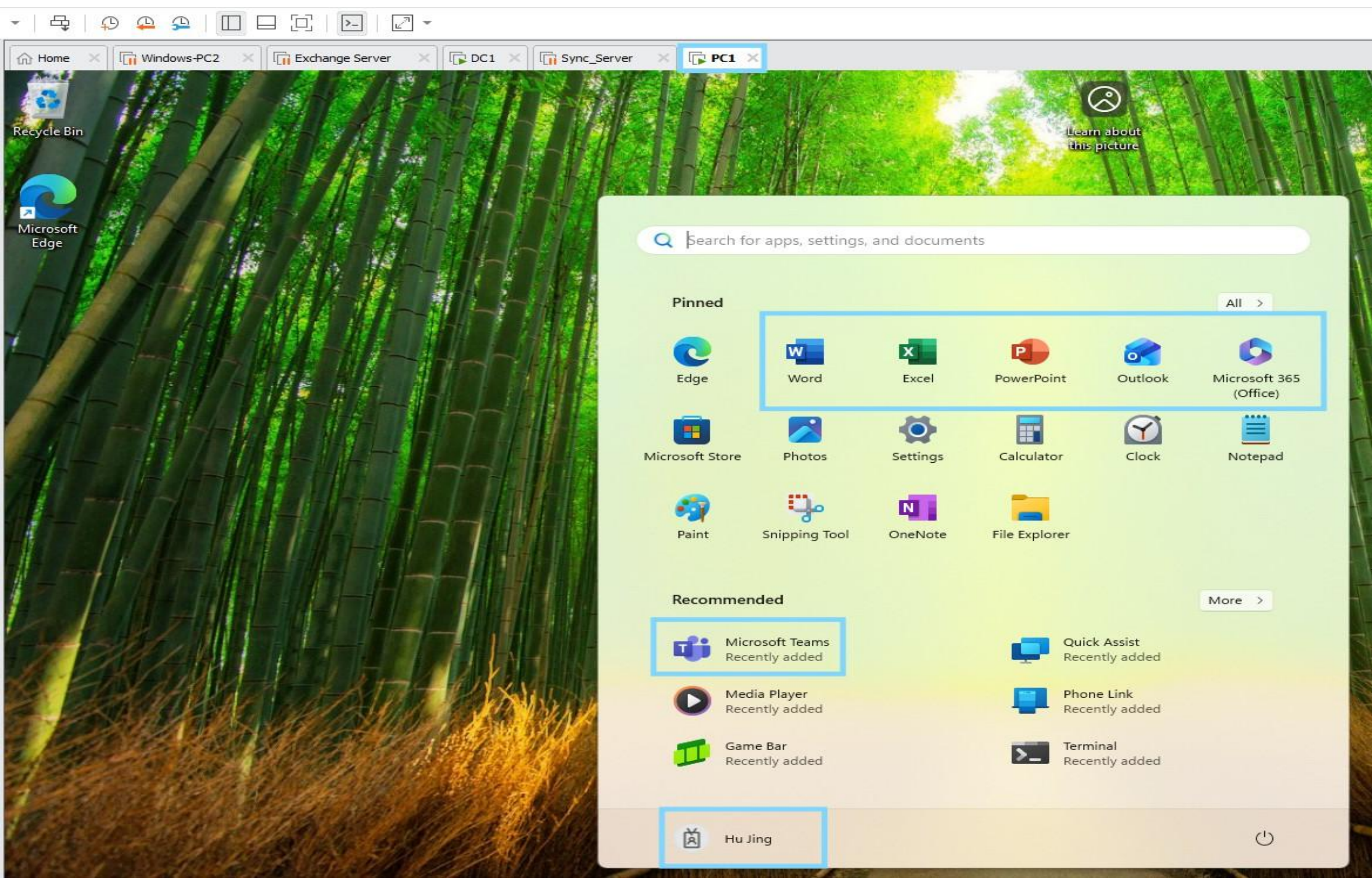
Windows Autopilot out-of-box experience where the user signs in with their Microsoft Entra ID work account to begin automated device enrollment and setup.



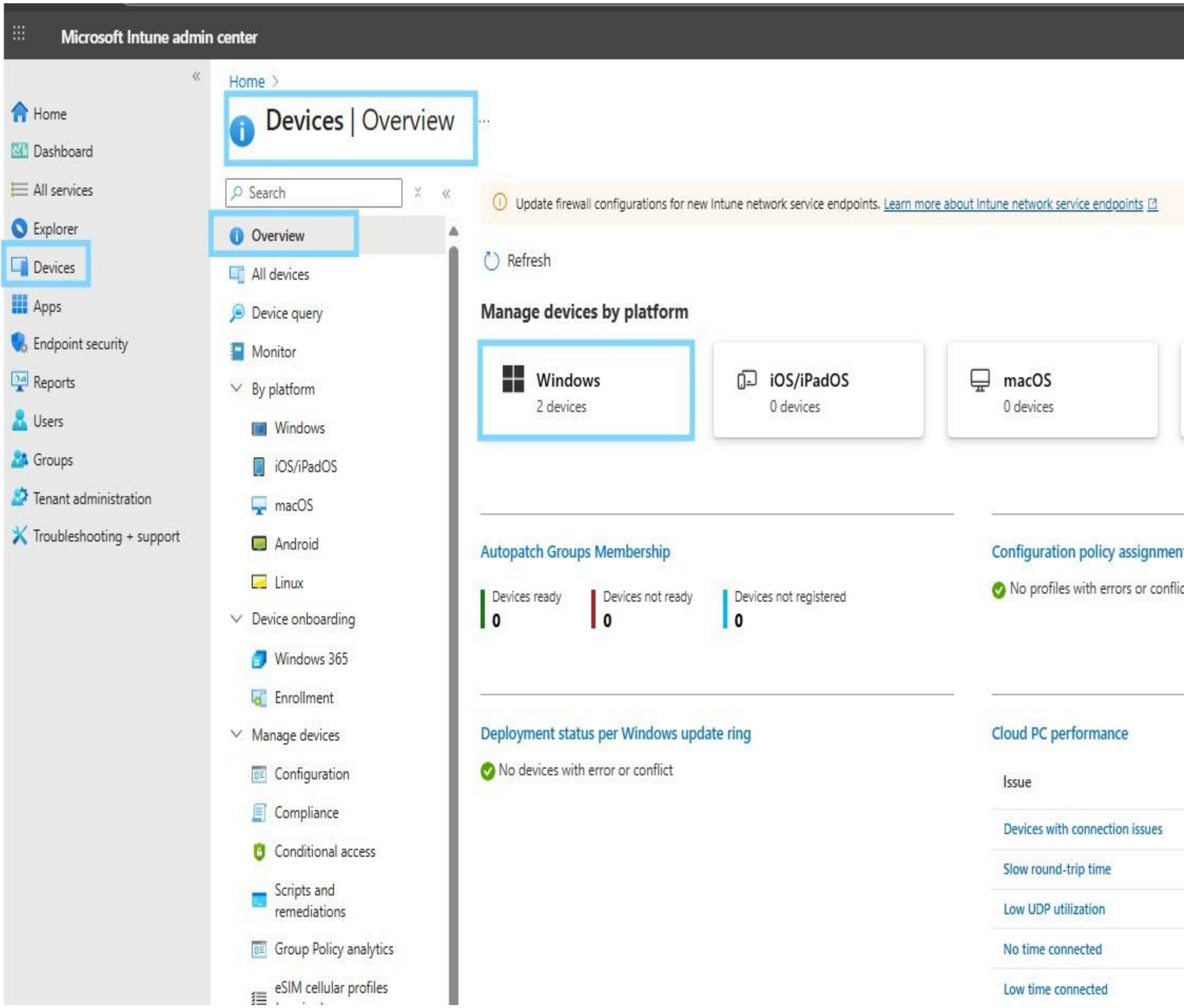
Windows Hello PIN setup during the Autopilot out-of-box experience to enable secure, password-less sign-in for the user's device.



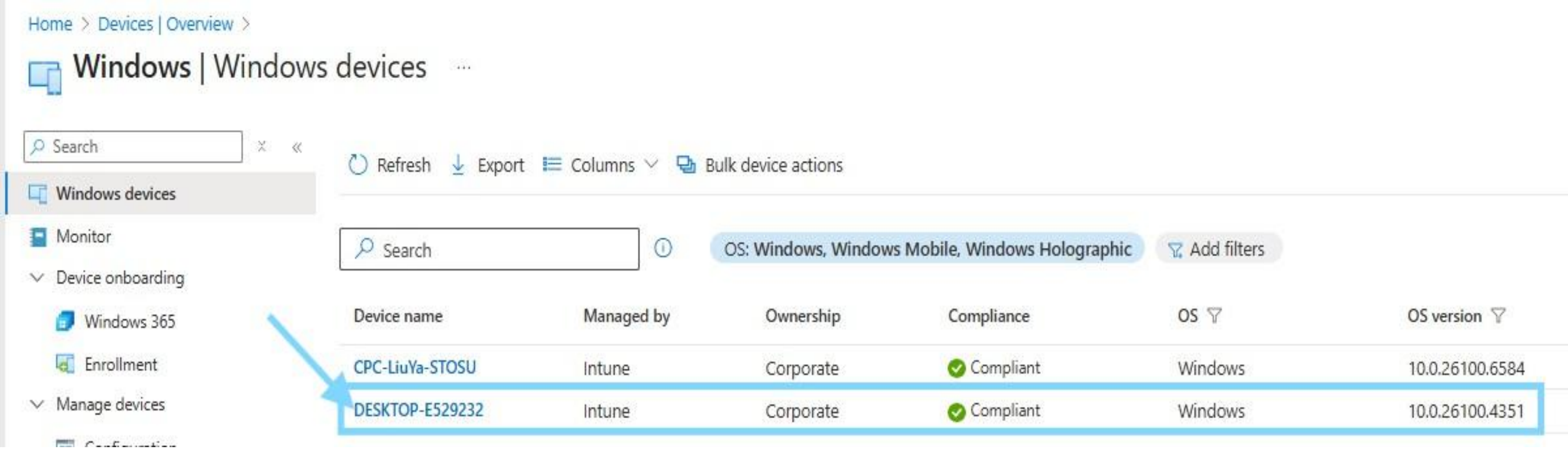
Windows Autopilot-provisioned device at first sign-in, with Microsoft 365 apps automatically installed and the user profile ready for use.



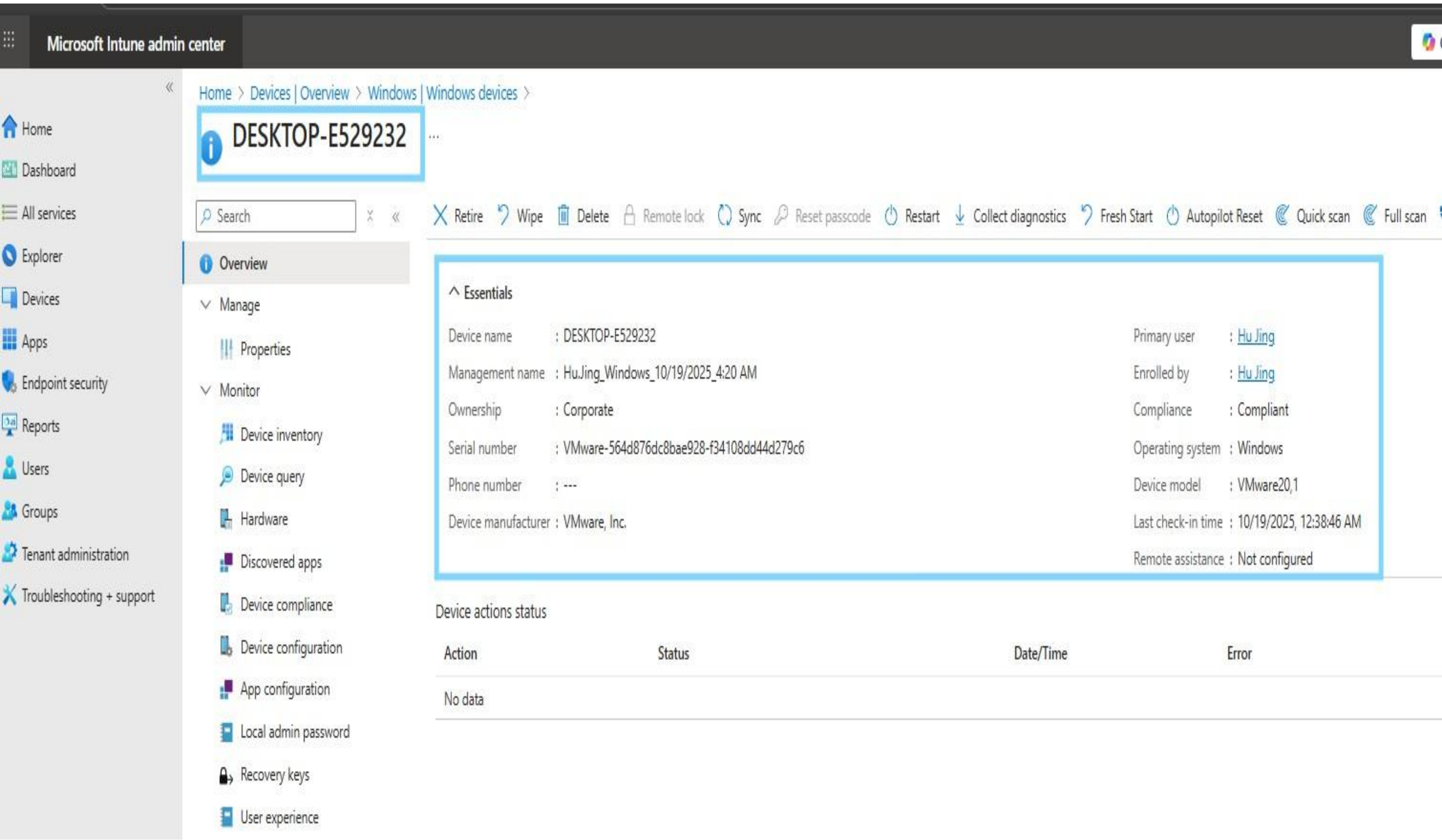
After setup is complete, the user will see Microsoft 365 apps like Word, Excel, PowerPoint, Outlook, and Teams in the Start menu. These apps are ready to use right away just click and start working.



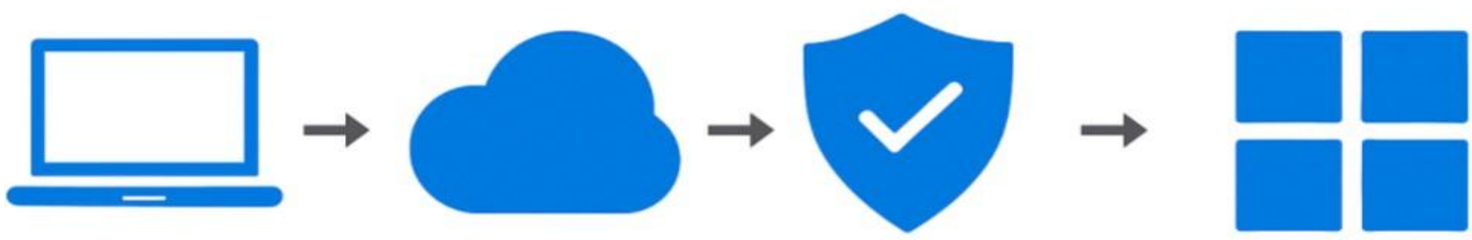
Windows devices successfully enrolled and managed by Microsoft Intune, marked as compliant and corporate-owned after Autopilot provisioning.

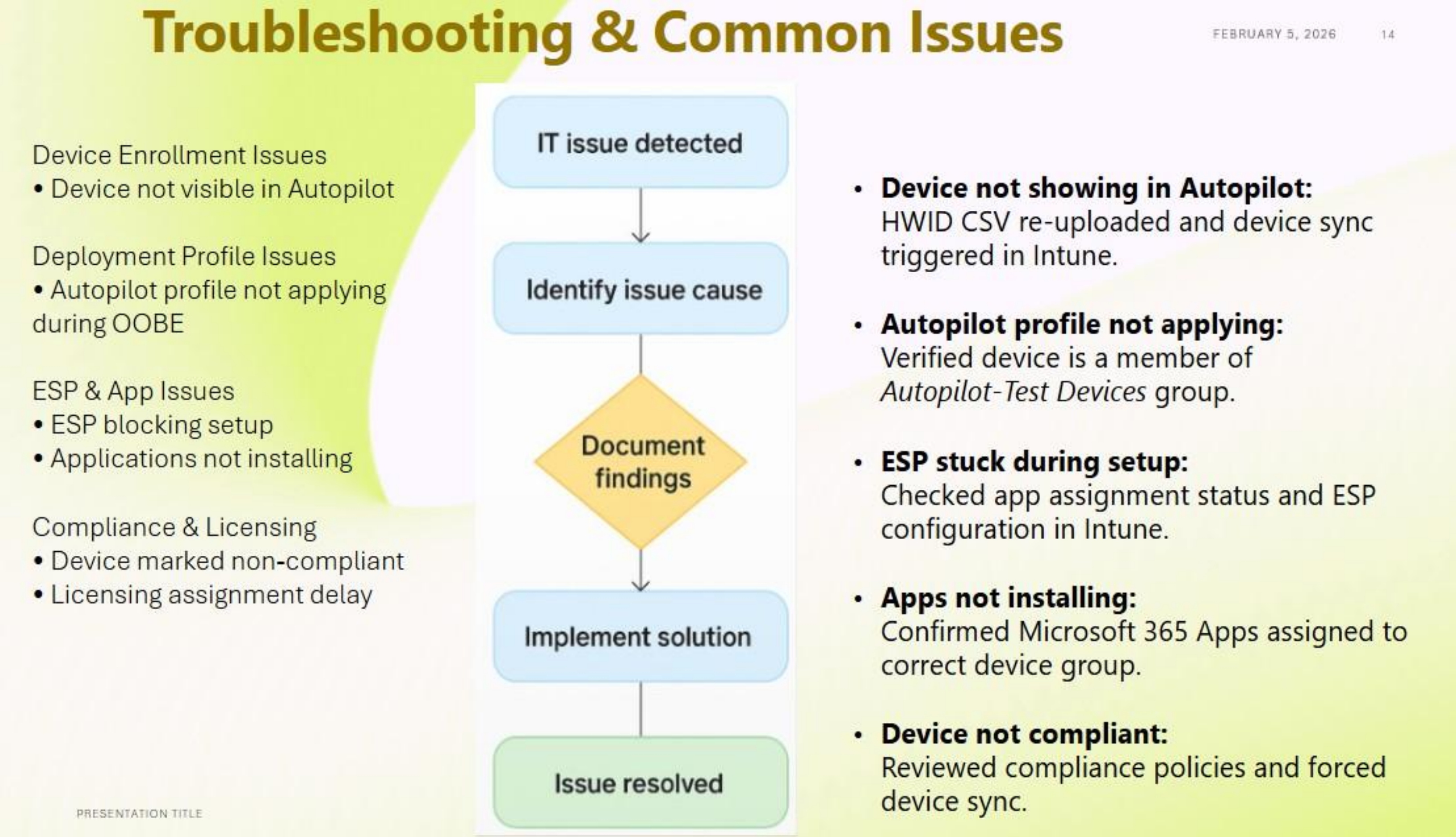


Detailed overview of a Windows device in Microsoft Intune, confirming it is enrolled, corporate-owned, compliant, and fully managed after Autopilot provisioning.



- This device is assigned and compliant in Intune. It belongs to user Hu Jing, runs Windows on a VMware virtual machine, and meets all company policies. The last check-in confirms it is active and ready for use.





- Gained hands-on experience with Windows Autopilot and Microsoft Intune
- Learned how zero-touch provisioning improves device deployment efficiency
- Understood real-world device lifecycle management in enterprise environments

✅ Conclusion: This project successfully demonstrated the end-to-end implementation of Windows Autopilot Device Enrolment using Microsoft Intune and Microsoft Entra ID. By capturing and importing the device hardware hash, configuring Autopilot deployment profiles, Enrolment Status Page (ESP), device groups, and application assignments, the device was fully provisioned with zero-touch deployment.

The solution ensured that the device was secure, compliant, and ready for use from the first sign-in, with Microsoft 365 applications and security policies applied automatically. This approach eliminated manual imaging, reduced IT workload, and provided a consistent onboarding experience, especially valuable for remote and hybrid work environments.
Overall, this implementation highlights how Windows Autopilot enables modern device management by improving security, efficiency, scalability, and user experience, making it an ideal solution for today’s cloud-first organizations. Hands-on experience delivering modern, secure endpoint management with Windows Autopilot, gaining valuable real-world IT experience along the way.